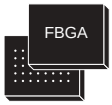
M69KB048BD
M69AB048BD

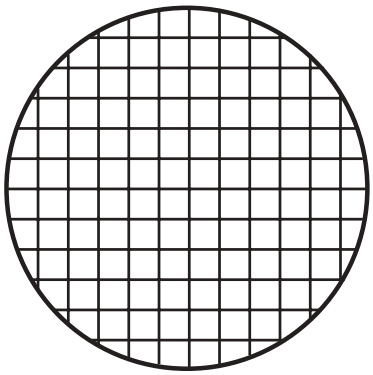
32 Mbit (2M x16) 1.8V Supply, Burst PSRAMs

Features

- Asynchronous SRAM Interface
- Fast Access Time: 70ns
- Asynchronous Page Read
  - Page Size: 8 Words
  - First Access within page: 70ns
  - Subsequent read within page: 20ns
- Synchronous Burst Read/Write Capability
  - Fixed Length (8, or 16 words) or Continuous mode
  - Output Delay: 8ns max
  - Burst Read/Write mode: fixed Length (8 or 16 Words) or Continuous
- Low Voltage Operating Condition
  - V_{CC} = 1.7 to 1.95V
- Byte Control By $\overline{UB}$ and $\overline{LB}$
- Low Power Consumption
  - Active Current: 30mA
  - Standby Current: 100µA
- Three Power-down Modes
  - Deep Power-Down
  - Partial Array Refresh of 4 Mbits
  - Partial Array Refresh of 8 Mbits
- Wide Operating Temperature
  - –30 to +85°C
- ECOPACK FBGA71 Package available



FBGA71 (ZA)
7 x 11mm



Wafer

Contents

1 Description 6

2 Signals description 9

2.1 Address inputs (A0-A20) 9
2.2 Data inputs/outputs (DQ8-DQ15) 9
2.3 Data inputs/outputs (DQ0-DQ7) 9
2.4 Chip Enable input (E1) 9
2.5 Chip Enable input (E2) 9
2.6 Output Enable input (G) 9
2.7 Write Enable input (W) 10
2.8 Upper Byte Enable input (UB) 10
2.9 Lower Byte Enable input (LB) 10
2.10 Clock input (K) 10
2.11 Latch Enable input (L) 10
2.12 Wait output (WAIT) 11
2.13 V_{CC} Supply Voltage 11
2.14 VSS Ground 11

3 Operation 14

3.1 Power-Up 14
3.2 Set Configuration register sequence 14
3.3 Power-Down modes 16
3.4 Asynchronous Page Read mode 17
3.5 Asynchronous Write mode 17
3.6 Synchronous Burst Read and Write modes 17
3.7 Synchronous Burst Read Suspend 18
3.8 Synchronous Burst Read Terminate 18
3.9 Synchronous Burst Write Suspend 19
3.10 Synchronous Burst Write Terminate 19

4 Maximum rating 24

5 DC and AC parameters 25

6 Package mechanical 60

7 Part numbering 61

8 Revision history 62

List of tables

Table 1. Signal names . . . 7
Table 2. Bus modes– Asynchronous mode . . . 12
Table 3. Bus modes– Synchronous Burst mode . . . 13
Table 4. Set CR Register sequence . . . 15
Table 5. Configuration Register definition . . . 15
Table 6. Data retention characteristics . . . 16
Table 7. Burst Length Definition . . . 19
Table 8. Absolute maximum ratings . . . 24
Table 9. Operating and AC measurement conditions . . . 25
Table 10. Capacitance . . . 26
Table 11. DC characteristics . . . 26
Table 12. Asynchronous Read AC characteristics (Page mode) . . . 28
Table 13. Asynchronous Write mode AC characteristics . . . 31
Table 14. Synchronous mode - Clock AC characteristics . . . 38
Table 15. Synchronous mode - Address Latch AC characteristics . . . 38
Table 16. Synchronous Burst Read AC characteristics . . . 39
Table 17. Synchronous Burst Write AC characteristics . . . 46
Table 18. Power-Down and Power-Up AC characteristics . . . 56
Table 19. Standby AC characteristics . . . 56
Table 20. FBGA71 7x11mm - 8x12 active ball array, 0.8mm pitch, package mechanical data . . . 60
Table 21. Ordering information scheme . . . 61
Table 22. Document revision history . . . 62

List of figures

Figure 1. Logic diagram . . . 7
Figure 2. FBGA Connections (Top view through package) . . . 8
Figure 3. Block diagram . . . 11
Figure 4. Synchronous Burst Read . . . 20
Figure 5. Synchronous Burst Write . . . 21
Figure 6. Burst Length Definition . . . 21
Figure 7. Latency definition . . . 22
Figure 8. Write Enable control modes . . . 23
Figure 9. AC measurement I/O waveform . . . 25
Figure 10. AC measurement load circuit . . . 26
Figure 11. Asynchronous Read Basic AC waveforms . . . 29
Figure 12. G controlled Asynchronous Read Basic AC waveforms . . . 29
Figure 13. LB and UB controlled Asynchronous Read basic AC waveforms . . . 30
Figure 14. E1 controlled, Asynchronous Page Read AC waveforms . . . 30
Figure 15. Asynchronous Write AC waveforms . . . 32
Figure 16. W controlled, Asynchronous Write AC waveforms . . . 33
Figure 17. W, UB and LB controlled, Asynchronous Write AC waveforms 1 . . . 33
Figure 18. W, UB and LB controlled, Asynchronous Write AC waveforms 2 . . . 34
Figure 19. W, UB and LB controlled, Asynchronous Write AC waveforms 3 . . . 34
Figure 20. W, UB and LB Controlled, Asynchronous Write AC Waveforms 4 . . . 35
Figure 21. E1 Controlled, Asynchronous Read Followed by Write AC Waveforms . . . 35
Figure 22. E1, W and G Controlled, Asynchronous Read Followed by Write AC Waveforms . . . 36
Figure 23. W and G Controlled, Asynchronous Read Followed by Write AC Waveforms . . . 36
Figure 24. W, G and UB/LB Controlled, Asynchronous Read Followed by Write AC Waveforms . . . 37
Figure 25. Clock Input Waveform . . . 40
Figure 26. Synchronous Burst Address Latch waveforms . . . 40
Figure 27. G controlled, Synchronous Read AC waveforms . . . 41
Figure 28. E1 controlled, Synchronous Read AC waveforms . . . 42
Figure 29. L controlled, Synchronous Read AC waveforms . . . 43
Figure 30. Synchronous Burst Read Suspend and Resume AC Waveforms . . . 44
Figure 31. Synchronous Burst Read Terminate AC waveforms . . . 45
Figure 32. W Level Control mode, Synchronous Burst Write AC waveforms . . . 47
Figure 33. W Pulse Control mode, Synchronous Burst Write AC waveforms . . . 48
Figure 34. L Controlled, Synchronous Write AC waveforms . . . 49
Figure 35. Synchronous Burst Write Suspend and Resume AC waveforms . . . 50
Figure 36. Synchronous Burst Write Terminate AC waveforms . . . 51
Figure 37. E1 controlled Synchronous Burst Read followed by Write AC waveforms . . . 52
Figure 38. L controlled Synchronous Burst Read followed by Write AC waveforms . . . 53
Figure 39. E1 controlled Synchronous Burst Write followed by Read AC waveforms . . . 54
Figure 40. L controlled Synchronous Burst Write followed by Read AC Waveforms . . . 55
Figure 41. Power-Up AC Waveforms - 1 . . . 56
Figure 42. Power-Up AC Waveforms - 2 . . . 57
Figure 43. Power Down Mode AC Waveforms . . . 57
Figure 44. Standby Mode Entry AC Waveforms, After Read . . . 57
Figure 45. Set Configuration Register Sequence (Asynchronous mode) AC Waveforms . . . 58
Figure 46. Set Configuration Register Sequence (Synchronous mode) AC Waveforms . . . 59
Figure 47. FBGA71 7x11mm - 8x12 active ball array, 0.8mm pitch, Package Outline, Bottom View . 60

1 Description

The M69KB048BD and M69AB048BD are 32 Mbit (33,554,432 bit) CMOS memories, organized as 2,097,152 words by 16 bits, and supplied by a single 1.7V to 1.95V supply voltage range.

The M69KB048BD and M69AB048BD are members of STMicroelectronics Pseudo SRAM (PSRAM) family whose memory array is implemented by using one transistor per cell topology in order to achieve bigger array sizes. The internal control logic of the M69KB048BD and M69AB048BD handles automatically the periodic refresh cycle, and without user involvement.

The devices support asynchronous page read mode and synchronous burst read and write modes for fast memory access.

They feature various kinds of Power-Down modes for power saving as a user configurable option:

- The Partial Array Refresh (PAR) performs a limited refresh of the part of the PSRAM array (4 Mbits, 8 Mbits) that contains essential data.
- The Deep Power-Down mode.

The M69AB048BD is available in a FBGA71 (0.8mm pitch) package. In order to meet environmental requirements, ST also offers the M69AB048BD in a FBGA71 ECOPACK® package. ECOPACK® packages have a Lead-free second level interconnect. The category of second Level Interconnect is marked on the package and on the inner box label, in compliance with JEDEC Standard JESD97. The maximum ratings related to soldering conditions are also marked on the inner box label. ECOPACK is an ST trademark. ECOPACK specifications are available at: www.st.com.

The M69KB048BD is offered in unsawn wafer.

The M69KB048BD and M69AB048BD are particularly suited for mobile applications such as Cellular Handsets, PDAs and handheld equipment.

Figure 1. Logic diagram

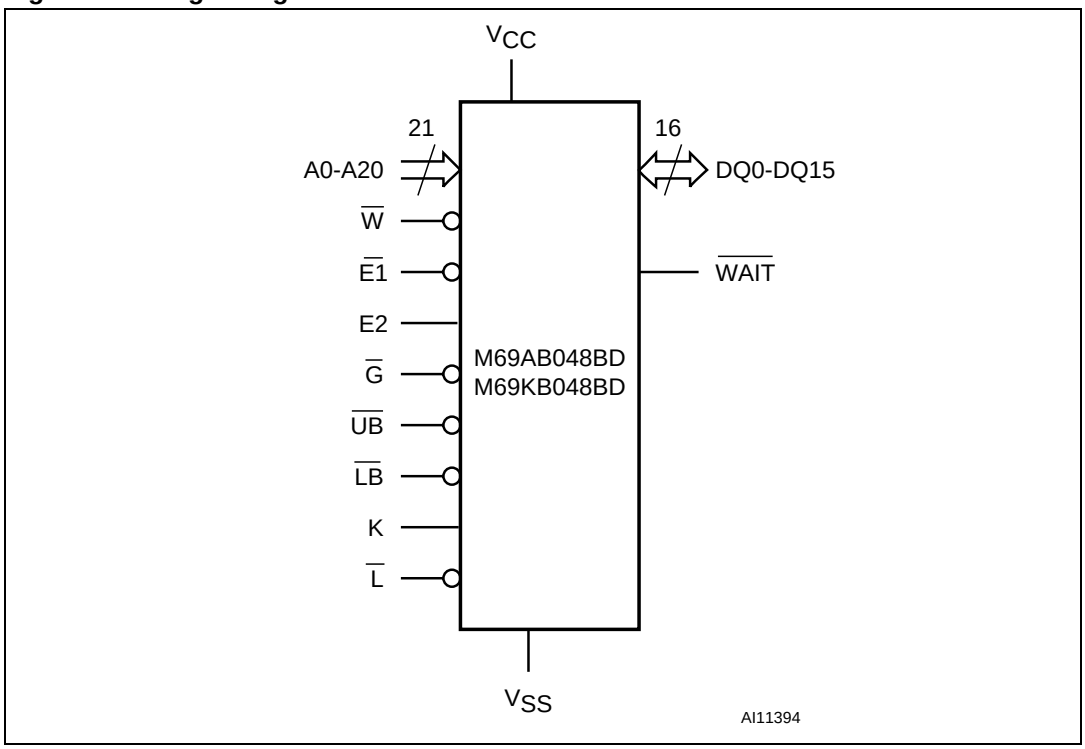
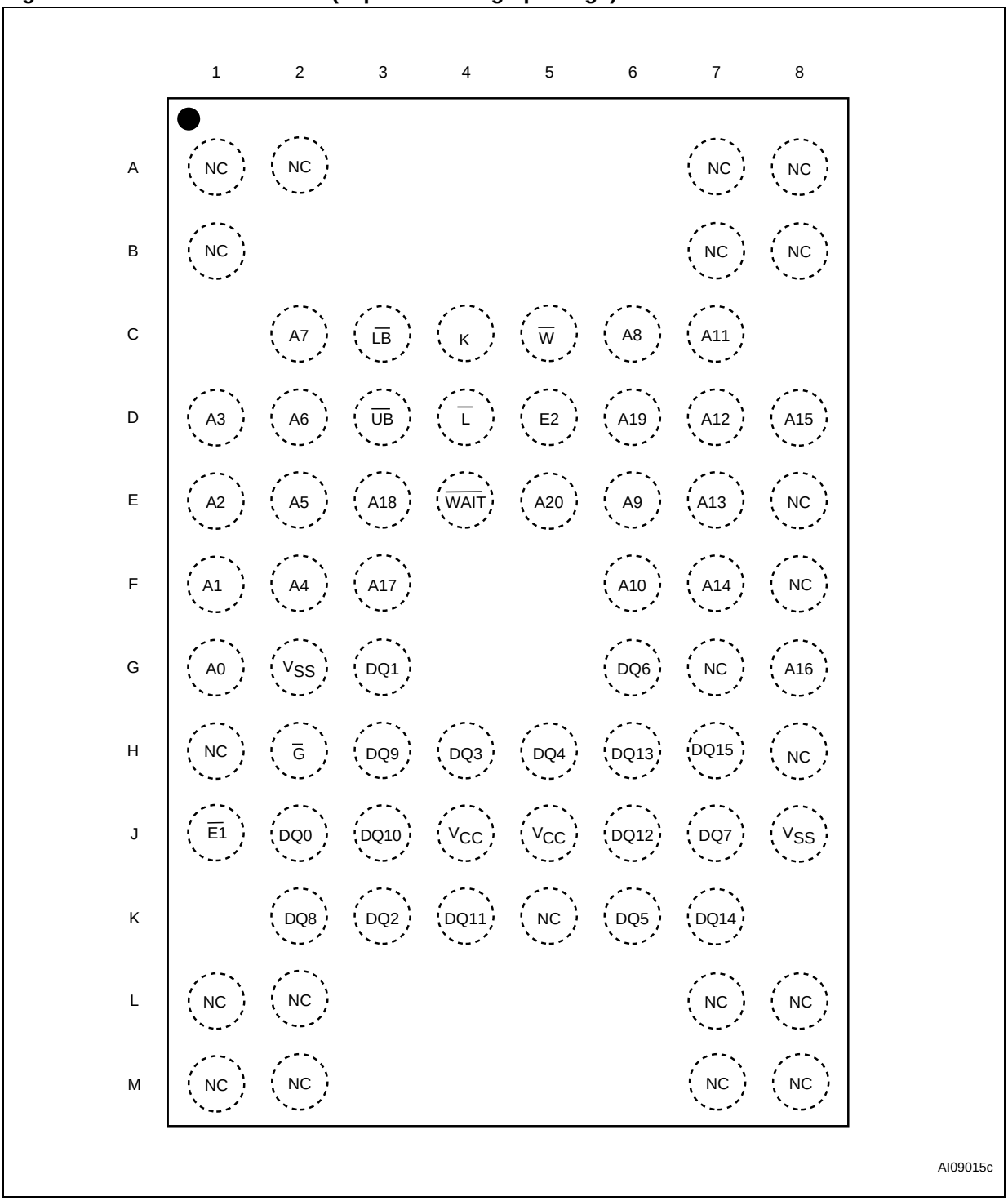


Table 1. Signal names

A0-A20	Address Inputs
DQ0-DQ15	Data Inputs/Outputs
$\overline{E1}$	Chip Enable Input (Active Low)
E2	Chip Enable Input (Active High)
$\overline{G}$	Output Enable Input
$\overline{W}$	Write Enable Input
$\overline{UB}$	Upper Byte Enable Input
$\overline{LB}$	Lower Byte Enable Input
K	Clock Input
$\overline{L}$	Latch Enable Input
$\overline{WAIT}$	Wait Output
V_{CC}	Core Supply Voltage
V_{SS}	Ground

Figure 2. FBGA Connections (Top view through package)



2 Signals description

See *Figure 1: Logic diagram*, *Figure 3: Block diagram*, and *Table 1: Signal names*, for a brief overview of the signals connected to this device.

2.1 Address inputs (A0-A20)

The Address Inputs select the cells in the memory array to access during Read and Write operations. These signals are also used during the Set Configuration Register sequence.

2.2 Data inputs/outputs (DQ8-DQ15)

The Upper Byte Data Inputs/Outputs carry the data to or from the upper part of the selected address during a Write or Read operation, when Upper Byte Enable ($\overline{UB}$) is driven Low.

2.3 Data inputs/outputs (DQ0-DQ7)

The Lower Byte Data Inputs/Outputs carry the data to or from the lower part of the selected address during a Write or Read operation, when Lower Byte Enable ($\overline{LB}$) is driven Low.

2.4 Chip Enable input ($\overline{E1}$)

When asserted (Low), the Chip Enable input, $\overline{E1}$, activates the memory state machine, address buffers and decoders, allowing Read and Write operations to be performed. When de-asserted (High), all other pins are ignored, and the device is automatically put, in Standby mode. The transition of $\overline{E1}$ from Low to High is also used to terminate a Synchronous Burst operation or during a Set Configuration Register sequence (see *Figure 31: Synchronous Burst Read Terminate AC waveforms*, and *Figure 36: Synchronous Burst Write Terminate AC waveforms*)

2.5 Chip Enable input (E2)

When de-asserted (Low), the Chip Enable input, E2, puts the device in Power-Down mode according to the Configuration Register settings (see *Table 5*).

2.6 Output Enable input ($\overline{G}$)

The Output Enable, $\overline{G}$, provides a high speed tri-state control, allowing fast read/write cycles to be achieved with the common I/O data bus. This signal is also used in Synchronous Burst Read mode.

2.7 Write Enable input ($\overline{W}$)

The Write Enable, $\overline{W}$, controls the Bus Write operation of the memory's Command Interface. This signal is also used in Synchronous Burst Read and Write mode.

Two types $\overline{W}$ control methods are available in Synchronous Burst Write mode:

- Level Control - The first Clock K rising edge following $\overline{W}$ falling edge triggers the Synchronous Burst Write.
- Pulse Control- A Write Enable, $\overline{W}$, Low pulse during a rising edge of the Clock K triggers the Synchronous Burst Write.

Refer to *Figure 8: Write Enable control modes* for details on the two $\overline{W}$ control modes.

The device is configured to operate in Level Control mode or Pulse Control mode through bit WC of the Configuration Register, CR (see *Table 5: Configuration Register definition*).

2.8 Upper Byte Enable input ($\overline{UB}$)

The Upper Byte Enable, $\overline{UB}$, gates the data on the Upper Byte Data Inputs/Outputs (DQ8-DQ15) to or from the upper part of the selected address during a Write or Read operation. This signal is only used during asynchronous operations and must be stable in Burst mode.

2.9 Lower Byte Enable input ($\overline{LB}$)

The Lower Byte Enable, $\overline{LB}$, gates the data on the Lower Byte Data Inputs/Outputs (DQ0-DQ7) to or from the lower part of the selected address during a Write or Read operation.

2.10 Clock input (K)

Clock, K, is an input signal to synchronize the memory to the microcontroller or system bus frequency during Synchronous Burst Read and Write operations. The Clock input increments the device internal address counter. The valid edge of K is the reference for latency counts from address latch, burst write data latch, and burst read data out. The K input is required during all synchronous operations, except in Standby and Power-Down. The Clock signal is "Don't care" during asynchronous operations.

2.11 Latch Enable input ($\overline{L}$)

The Latch Enable input is used in both asynchronous and synchronous mode to validate the address present on A0-A20. All the addresses are latched on the rising edge of $\overline{L}$.

$\overline{L}$ input is active after $\overline{E1}$ becomes Low, V_{IL}, and is disabled when $\overline{E1}$ is High, V_{IH}.

The address latch is transparent when $\overline{L}$ is at V_{IL} and it is inhibited when $\overline{L}$ is at V_{IH}. Once the first address has been latched, the state of $\overline{L}$ controls whether subsequent addresses come from the address lines ($\overline{L}$=V_{IL}) or from the internal Burst counter ($\overline{L}$=V_{IH}).

2.12 Wait output ($\overline{WAIT}$)

$\overline{WAIT}$ is an output signal used during Synchronous Burst Read operations. It indicates whether the data on the output bus are valid. In Synchronous Read Suspend mode, the $\overline{WAIT}$ remains in its previous state.

The $\overline{WAIT}$ signal is gated by $\overline{G}$ and is asserted after a specified delay from the falling edge of $\overline{G}$. It becomes High one clock cycle before valid data is output.

When the device is not in Synchronous Burst Read mode, $\overline{WAIT}$ is high impedance.

2.13 V_{CC} Supply Voltage

The V_{CC} Supply Voltage supplies the power for all operations (Read, Write, etc.) and for driving the refresh logic, even when the device is not being accessed.

2.14 V_{SS} Ground

The V_{SS} Ground is the reference for all voltage measurements.

Figure 3. Block diagram

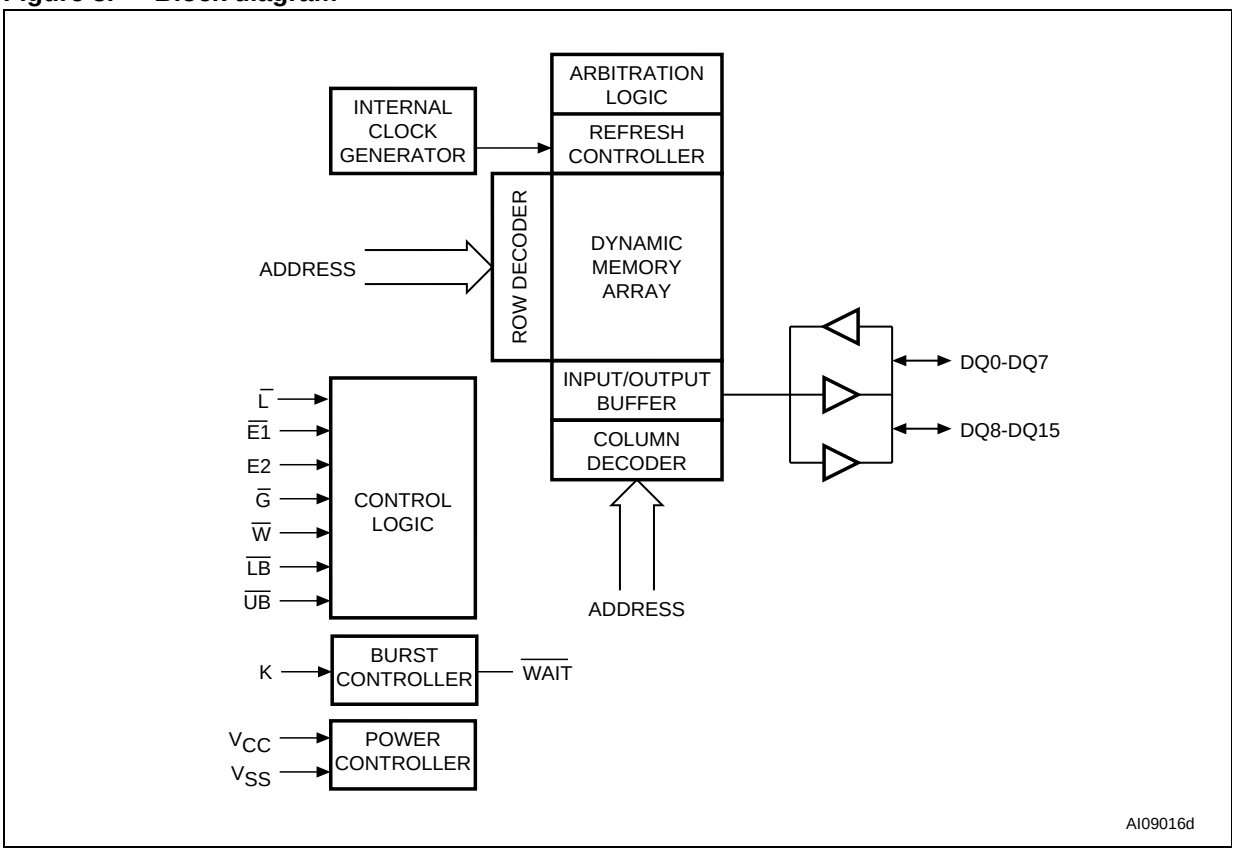


Table 2. Bus modes– Asynchronous mode

Mode	E2	$\overline{E1}$	K	$\overline{L}$	$\overline{W}$	$\overline{G}$	$\overline{LB}$	$\overline{UB}$	A0-A20	DQ0-DQ7	DQ8-DQ15	$\overline{WAIT}$
Standby (Deselected)	V_{IH}	V_{IH}	X	X	X	X	X	X	X	High-Z	High-Z	High-Z
Output Disable(1)	V_{IH}	V_{IL}	X	(2)	V_{IH}	V_{IH}	X	X	(3)	High-Z	High-Z	High-Z
Output Disable (No Read)	V_{IH}	V_{IL}	X	(2)	V_{IH}	V_{IL}	V_{IH}	V_{IH}	Valid	High-Z	High-Z	High-Z
Upper Byte Read	V_{IH}	V_{IL}	X	(2)	V_{IH}	V_{IL}	V_{IH}	V_{IL}	Valid	High-Z	Output Valid	High-Z
Lower Byte Read	V_{IH}	V_{IL}	X	(2)	V_{IH}	V_{IL}	V_{IL}	V_{IH}	Valid	Output Valid	High-Z	High-Z
Word Read	V_{IH}	V_{IL}	X	(2)	V_{IH}	V_{IL}	V_{IL}	V_{IL}	Valid	Output Valid	Output Valid	High-Z
Page Read	V_{IH}	V_{IL}	X	X	V_{IH}	V_{IL}	V_{IL}/V_{IH}	V_{IL}/V_{IH}	Valid	(4)	(4)	High-Z
No Write	V_{IH}	V_{IL}	X	(2)	V_{IL}	V_{IH}(5)	V_{IH}	V_{IH}	Valid	Invalid	Invalid	High-Z
Upper Byte Write	V_{IH}	V_{IL}	X	(2)	V_{IL}	V_{IH}(5)	V_{IH}	V_{IL}	Valid	Invalid	Input Valid	High-Z
Lower Byte Write	V_{IH}	V_{IL}	X	(2)	V_{IL}	V_{IH}(5)	V_{IL}	V_{IH}	Valid	Input Valid	Invalid	High-Z
Word Write	V_{IH}	V_{IL}	X	(2)	V_{IL}	V_{IH}(5)	V_{IL}	V_{IL}	Valid	Input Valid	Input Valid	High-Z
Power-Down(6)	V_{IL}	X	X	X	X	X	X	X	X	High-Z	High-Z	High-Z

1. The device must not be kept in the Output Disable state during more than 1µs.
2. The address latch is transparent when $\overline{L}$ is at V_{IL}. The addresses are latched on the rising edge of $\overline{L}$.
3. Address bits are either at V_{IL} or V_{IH} but must be valid before Read or Write operation.
4. Data outputs are either valid or Hi-Z depending on the state of $\overline{LB}$/$\overline{UB}$ inputs.
5. $\overline{G}$ can be V_{IL} during a Write operation if the following conditions are satisfied:
   a. Write pulse is initiated by $\overline{E1}$, or cycle time of the previous operation cycle is satisfied;
   b. $\overline{G}$ stays V_{IL} during the entire Write cycle.
6. Power-Down mode can be entered from Standby state and all DQ pins are in High-Z state. Data retention depends on the Power-Down mode selected.

Table 3. Bus modes– Synchronous Burst mode

Mode	E2	$\overline{E1}$	K	$\overline{L}$	$\overline{W}$	$\overline{G}$	$\overline{LB}$	$\overline{UB}$	A0-A20	DQ0-DQ7	DQ8-DQ15	$\overline{WAIT}$
Standby (Deselected)	V_{IH}	V_{IH}	X	X	X	X	X	X	X	High-Z	High-Z	High-Z
Initial Burst Read/Write(1)	V_{IH}	V_{IL}	_/‾(2)	‾_/‾ ‾↑‾(3)	X(4)	X(4)	X(5)	X(5)	Valid (6)	High-Z(7)	High-Z(7)	High-Z(8)
Advance Burst Read(1)	V_{IH}	V_{IL}	_/‾(2)	V_{IH}	V_{IH}	V_{IL}	X(5)	X(5)	X	Output Valid(9)	Output Valid(9)	Output Valid
Burst Read Suspend(1)	V_{IH}	V_{IL}	_/‾(2)	V_{IH}	V_{IH}	V_{IH}	X(5)	X(5)	X	High-Z	High-Z	High-Z(10)
Terminate Burst Read	V_{IH}	_/‾	X	V_{IH}	V_{IH}	V_{IL}	X(5)	X(5)	X	High-Z	High-Z	High-Z
Advance Burst Write(1)	V_{IH}	V_{IL}	_/‾(2)	V_{IH}	V_{IL} (13)	V_{IH}	X(5)	X(5)	X	Input Valid(11)	Input Valid(11)	V_{IH}(12)
Burst Write Suspend(1)	V_{IH}	V_{IL}	_/‾(2)	V_{IH}	V_{IH} (13)	V_{IH}	X(5)	X(5)	X	Input Invalid	Input Invalid	High-Z(10)
Terminate Burst Write	V_{IH}	_/‾	X	V_{IH}	X	V_{IH}	X(5)	X(5)	X	High-Z	High-Z	High-Z
Power-Down(14)	V_{IL}	X	X	X	X	X	X	X	X	High-Z	High-Z	High-Z

1. The device must not be kept in the Output Disable state during more than 4µs.
2. The valid edge of the Clock signal is configured through the Valid Clock Edge bit (VE) of the Configuration Register. The Clock signal must be stable before accessing the memory array.
3. The Address Valid input latches the address bits on its rising edge.
4. $\overline{W}$ and $\overline{G}$ can be either at V_{IL} or V_{IH} but must not be both at V_{IL} (prohibited configuration).
5. $\overline{LB}$ and $\overline{UB}$ can be either at V_{IL} or V_{IH} but must be valid before Synchronous Read or Write operation. Once $\overline{LB}$ and $\overline{UB}$ are set, they must remain stable until the end of the Synchronous Burst operation.
6. Once the address is latched, address bits must remain stable while $\overline{L}$ is Low, V_{IL}.
7. If $\overline{G}$ is at V_{IL}, data outputs are either Invalid or Hi-Z depending on the state of $\overline{LB}$ and $\overline{UB}$ inputs. If $\overline{W}$ is at V_{IL}, data input are Invalid. If $\overline{W}$ and $\overline{G}$ are at V_{IH}, data outputs are Hi-Z.
8. Data outputs are either valid or invalid depending on the state of $\overline{G}$ and $\overline{W}$ inputs.
9. Data outputs are either valid or Hi-Z depending on the state of $\overline{LB}$ and $\overline{UB}$ inputs.
10. Keep the state of previous cycle.
11. Data input are either valid or invalid depending on the state of $\overline{LB}$ and $\overline{UB}$ inputs.
12. WAIT remains High, V_{IH}, during Advanced Burst Write operations.
13. When the device is operating in $\overline{W}$ Pulse Control mode, $\overline{W}$ is don't care once the Burst Write operation has been triggered by a $\overline{W}$ Low Pulse together with address latching. Burst Write Suspend is not supported in $\overline{W}$ Pulse Control mode.
14. Power-Down mode can be entered from Standby state and all DQ pins are in High-Z state. Data retention depends on the Power-Down mode selected.

3 Operation

This devices support Asynchronous Page Read, Page Read and Synchronous Burst Read and Write modes together with three kinds of power down modes for power saving.

3.1 Power-Up

The internal control logic of the M69KB048BD and M69AB048B needs to be initialized. The following power-up procedure must be followed before the memory is used:

1. Apply power and attempt to maintain $\overline{E1}$=E2=V_{IH}
2. Maintain stable power supply and $\overline{E1}$=E2=V_{IH} for t_{EHEV}.

After Power-up, the device is configured in Asynchronous Page Read, Asynchronous Write and Deep Power-Down mode (default settings of the CR Register).

See *Figure 42: Power-Up AC Waveforms - 2*, for details on the Power-up timing.

3.2 Set Configuration register sequence

The Configuration Register (CR) is used to define the device operating mode. The CR register is automatically loaded with default settings during power-up. The CR register can be loaded by latching the values placed on address lines A0 to A20 into the register.

The device configuration can be updated any time by issuing a Set Configuration Register sequence requiring 6 read/write operations to a unique address. This sequence can be issued both in Asynchronous and Synchronous mode (see *Figure 45* and *Figure 46* Set Configuration Register AC Waveforms).

The first cycle reads from address 1FFFFFh.

The second and third cycles write back to address 1FFFFFh the same data (RDa) read during the first cycle. If the second or third cycle writes to a different address, the Set Configuration Register sequence is cancelled and the cycles operate as normal write operations in the memory array.

The fourth and fifth cycles write to address 1FFFFFh. The data written during the fourth and fifth cycle is ‘don’t-care’. If the fourth and fifth cycles writes to a different address, the Set Configuration Register sequence is cancelled and the cycles operate as normal write operations in the memory array.

The sixth cycle effectively programs the content of the CR register by performing a Read operation with A0-A20 containing the CR value.

Between each Read or Write operation, the device must be put in Standby mode by holding $\overline{E1}$ High.

Once updated, the value of the CR register may be lost. A CR Set sequence should be performed prior to regular read/write operation if a device configuration different from the default one is requested.

The definition of the Configuration Register bits is detailed in *Table 5: Configuration Register definition*.

Table 4. Set CR Register sequence

Cycle	Operation	Address	Data
1st	Read	1FFFFFh	RDa(1)
2nd	Write	1FFFFFh	RDa(1)
3rd	Write	1FFFFFh	RDa(1)
4th	Write	1FFFFFh	X
5th	Write	1FFFFFh	X
6th	Read	CR(2)	RDb

1. RD Read Data.
2. CR Configuration Register value.

Table 5. Configuration Register definition

Address Bits	CR Bits	Definition	Value	Description
A20-A19	PS	Partial Size	00	8 Mbits Partial Array Refresh(1)
			01	4 Mbits Partial Array Refresh(1)
			10	Reserved for future use(2)
			11	Deep Power-Down mode (Default)
A18-A16	BL	Burst Length	010	8 words
			011	16 words
			111	Continuous
			The other configurations are reserved for future use (2)	
A15	M	Mode	0	Synchronous Burst Read/Write(3)
			1	Asynchronous (Page Read/Normal Write) (default)(3)
A14-A12	RL	Read Latency	001	3 clock cycles
			010	4 clock cycles
			011	5 clock cycles
			100	6 clock cycles
			Other configuration is reserved for future use(2)	
A11	—	—	1	Unused bit(4)
A10	SW	Single Write	0	Burst Read & Burst Write
			1	Burst Read & Single Write(5)
A9	VE	Valid Clock Edge	0	Reserved for future use (2)
			1	Rising Clock Edge
A8	RP	Reset to Page	0	Reset to the device to Page mode(6)
			1	Device remains in the previously selected mode(1)
A7	WC	Write Enable Control	0	$\overline{W}$ Pulse Control without Write Suspend Function(5)
			1	$\overline{W}$ Level Control with Write Suspend Function

Table 5. Configuration Register definition (continued)

Address Bits	CR Bits	Definition	Value	Description
A6	DS	Driver Strength Bit	0	Full Strength
			1	Half Strength (default)
A5-A0	—	—	111111	Unused bits(4)

1. Deep Power-Down and Partial Array Refresh are effective when RP bit is set to ‘1’.
2. This value is prohibited.
3. If the M bit is set to ‘0’ (Burst mode selected), all Burst mode related CR bits must be initialized simultaneously with appropriate values. If it is set to ‘1’, PS and DS bits must be initialized simultaneously with appropriate value while all the other CR bits are set to ‘1’.
4. A11 and A5 to A0 must be set to ‘1’.
5. Burst Read and Single Write are not supported in $\overline{W}$ Pulse Control mode.
6. If RP bit is set to ‘0’, driving E2 Low, V_{IL}, returns the device to Standby and asynchronous mode, regardless of the value of PS bits. Deep Power-Down and Partial Array Refresh are consequently unavailable.

3.3 Power-Down modes

Power-Down modes are used to put the device in low power idle state when E2 is Low.

This device has three power down modes:

- Partial Array Refresh (PAR) on 4 Mbit areas of the memory array
- Partial Array Refresh (PAR) on 8 Mbit areas of the memory array
- Deep Power-Down mode

The Power-Down mode is selected by issuing a Set CR sequence (PS bits).

In Deep Power-Down mode, all data store in the memory array are lost. In PAR mode, a limited refresh of part of the memory array (4 or 8 Mbit) is performed). The data retention characteristics of each Power-Down mode are described in *Table 18*.

By default, the Power-Down mode is “Deep Power-Down” which corresponds to the lowest power consumption.

Table 6. Data retention characteristics

Mode	Data Retention Size	Retention Address
Deep Power-Down (Default)	No	N/A
4 Mbit PAR	4 Mbits	00000h to 3FFFFh
8 Mit PAR	8 Mbits	00000h to 7FFFFh

3.4 Asynchronous Page Read mode

The Asynchronous Page read mode gives greater performance, even than the traditional Asynchronous Random Read mode. The page mode is not available for write operations.

By default, the device is configured in Asynchronous Page Read mode. $\overline{L}$ must be driven Low, V_{IL}, during all Asynchronous Page Read operations.

In Asynchronous Page Read mode, a Page of data is internally read. Each memory page consists of 8 Words, and has the same set of values on A3-A20; only of A0 to A2 differ. The first read operation within the Page has the normal access time (t_{AVQV}), subsequent reads within the same Page have much shorter access times (t_{AVQV2}). If the Page changes then the normal, longer timings apply again.

See *Figure 14* and *Table 12* for details of the Asynchronous Page Read timing requirements.

3.5 Asynchronous Write mode

Asynchronous Write operations occur when:

- Write Enable $\overline{W}$ is Low and
- Output Enable $\overline{G}$ is High and
- the two Chip Enable signals are asserted ($\overline{E1}$ Low, and E2 High) and
- at least one of the two signals $\overline{UB}$ or $\overline{LB}$ is Low.

The Asynchronous Write cycle begins t_{AVWL}, t_{AVEL} or t_{AVBL} after the address becomes valid depending on which of the above signals is the last to go Low.

Data inputs must be valid for t_{DVWH} before the rising edge of $\overline{W}$, or t_{DVEH} before the rising edge of $\overline{E1}$, or t_{DVBH} before the rising edge of $\overline{UB/LB}$ whichever occurs first, and remain valid for t_{WHDZ}, t_{EHDZ} or t_{BHDZ}.

The Asynchronous Write cycle is terminated by the first rising edge on $\overline{W}$ or $\overline{E1}$ signals.

See *Table 13* and *Figure 15* to *Figure 20* for details of Asynchronous Write AC timing requirements.

3.6 Synchronous Burst Read and Write modes

Burst mode allows high-speed synchronous read and write operations.

In Synchronous Burst modes the data is input or output to or from the memory array in bursts that are synchronized with the microcontroller or system clock.

This mode uses K, $\overline{L}$, and $\overline{WAIT}$ control signals (see SIGNAL DESCRIPTIONS section). In Synchronous Burst mode, the address is latched on the rising edge of $\overline{L}$ when $\overline{E1}$ is Low.

The minimum values of t_{LLLH}, duration of $\overline{L}$ Low pulse and t_{LLKH}, delay from $\overline{L}$ falling edge to the rising edge of the Clock K that starts the Burst Address Counter, must be respected.

The address becomes valid t_{AVLL} or t_{AVEL} after the falling edge of $\overline{L}$ or $\overline{E1}$ whichever occurs the last. The valid address must remain unchanged while $\overline{L}$ is Low.

See *Table 16* and *Figure 4* for details on Synchronous Burst Read Characteristics and Waveforms and *Table 17* and *Figure 5* for details on Synchronous Burst Write Characteristics and Waveforms.

After power-up, a Set Configuration Register (CR) sequence must be issued to put the device in Synchronous Burst mode:

- Mode bit, M, set '1' puts the device in Synchronous Burst mode.
- Read Latency bits, RL, defines the number of clock cycles between the address being latched and the first read data transferred. It can be set to 3, 4, 5, or 6 clock cycles. If the Read Latency has been configured to 'n', the Write Latency, that is the number of clock cycles between the address being latched and first write data transferred, is automatically set to 'n-1' (see *Figure 7: Latency definition*).
- Burst Length bits, BL, is the number of Words to be input or output during the Synchronous Burst Read or Write operation. It can be set to fixed length (8 Words or 16 Words) or Continuous. In Synchronous Burst mode, the internal Burst Address Counter is incremented at each Clock K rising edge starting from the initial address being latched, until the end of boundary address is reached and then wraps as shown in *Table 7*. When the number of Words defined by the BL bits has been input or output, the synchronous operation automatically ends. When the Burst Length is set to Continuous, Synchronous Burst Read or Write is endless unless a rising edge on $\overline{E1}$ occurs. During Continuous Burst Read operations, an additional output delay may occur when a Burst sequence crosses the row boundary. This is the case when the A0-A6 bits of the starting address are either '7Dh', '7Eh', or '7Fh' as shown in *Table 7: Burst Length Definition*. In this case, the $\overline{WAIT}$ signal is asserted.

3.7 Synchronous Burst Read Suspend

Synchronous Burst Read operations can be suspended by driving the Output enable input, $\overline{G}$, to High.

The first Clock K rising edge following $\overline{G}$ rising edge triggers the Synchronous Burst Read Suspend. The device internal address counter is suspended and data outputs become high impedance t_{GHQZ} after $\overline{G}$ rising edge. It is prohibited to suspend the first data output at the beginning of a Synchronous Burst Read.

During Synchronous Burst Read Suspend, the $\overline{WAIT}$ output signal remains unchanged from the cycle where data were suspended. In this case, it does not indicate data status.

Synchronous Burst Read operations can be resumed by driving $\overline{G}$ Low. Data outputs become valid and the device internal address counter is reactivated with a specific delay after $\overline{G}$ falling edge. The last data output being suspended as the result of $\overline{G}$ going High and the first data output after $\overline{G}$ returns to Low are read from the same memory address.

3.8 Synchronous Burst Read Terminate

Synchronous Burst Read operations can be terminated by driving $\overline{E1}$ to High. If the Burst Length is set to Continuous, Synchronous Burst Read operations goes on endlessly unless a rising edge on $\overline{E1}$ occurs. It is prohibited to terminate a Synchronous Burst Read before the first data is output. In order to ensure the validity of the last data output, a minimum duration of $\overline{E1}$ Low pulse from the rising edge of the Clock K is required.

3.9 Synchronous Burst Write Suspend

Synchronous Burst Write operations can be suspended by driving the Write enable input, $\overline{W}$, to High.

The first Clock K rising edge following $\overline{W}$ rising edge triggers the Synchronous Burst Write Suspend. The device internal address counter is suspended and data inputs become high impedance t_{WHQZ} after $\overline{W}$ rising edge. It is prohibited to suspend the first data input at the beginning of a Synchronous Burst Write.

Synchronous Burst Write operations can be resumed by setting $\overline{W}$ Low. Once $\overline{W}$ is driven Low, data inputs become valid after a specific delay, and internal address counter is reactivated.

After $\overline{W}$ returns Low, write operation restart from the address at which the Synchronous Burst Write Suspend occurred.

Synchronous Burst Write Suspend is available only when the device is operating in Write Enable Level Control mode.

3.10 Synchronous Burst Write Terminate

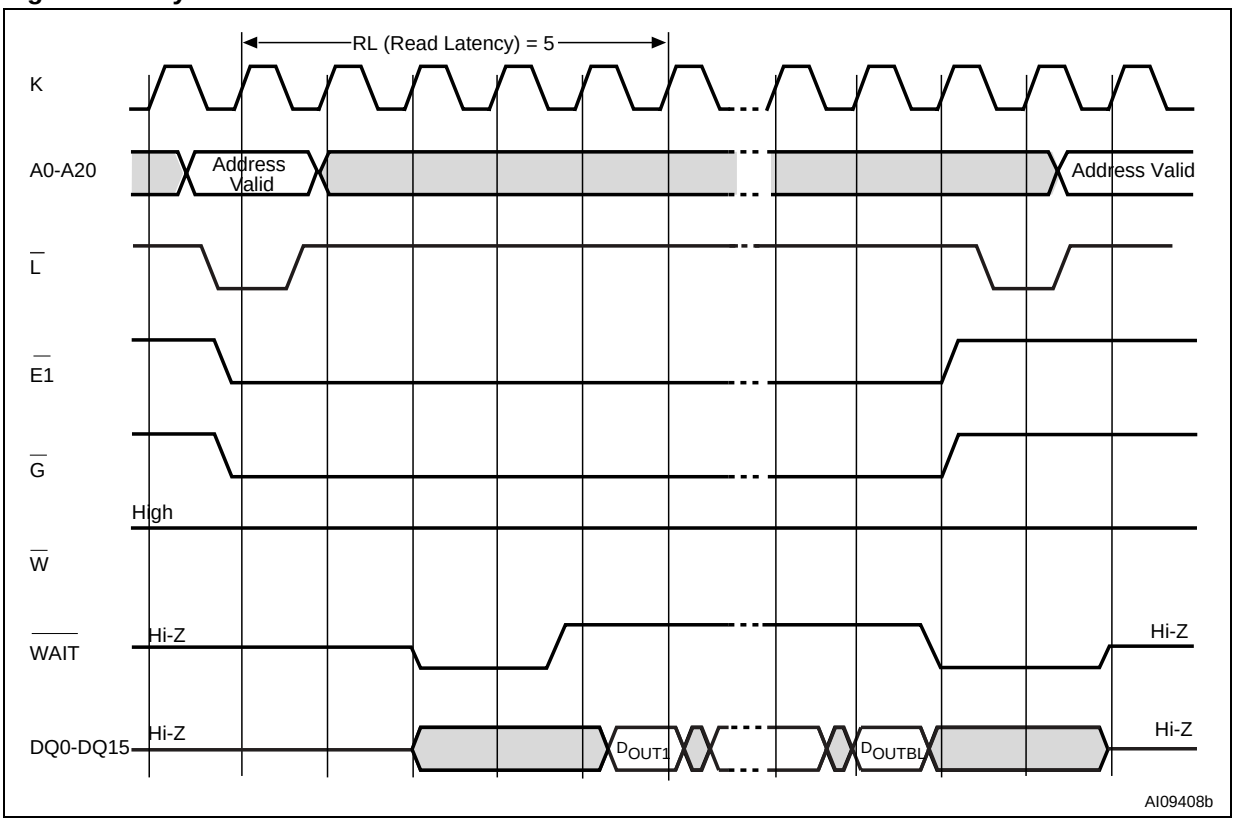
Synchronous Burst Write operations can be terminated by driving $\overline{E1}$ to High. If the Burst Length is set to Continuous, Synchronous Burst Write operations goes on endlessly unless a rising edge on $\overline{E1}$ occurs. It is prohibited to terminate a Synchronous Burst Write before the first data is latched. In order to ensure the validity of the last data write, a minimum duration of $\overline{E1}$ Low pulse from the rising edge of the Clock K is required.

Table 7. Burst Length Definition

Starting Address (A0-A6)(1)	8 Words	16 Words	Continuous
00h	00-01-02-...-06-07	00-01-02-...-0E-0F	00-01-02-03-04-...
01h	01-02-03-...-07-00	01-02-03-...-0F-00	01-02-03-04-05-...
02h	02-03-...-07-00-01	02-03-...-0F-00-01	02-03-04-05-06-...
03h	03-...-07-00-01-02	03-...-0F-00-01-02	03-04-05-06-07-...
...	...	...	...
7Ch	7C-...-7F-78-...-7B	7C-...-7F-70-...-7B	7C-7D-7E-7F-80-81-...
7Dh	7D-7E-7F-78-...-7C	7D-7E-7F-70-...-7C	7D-7E-7F-WAIT-80-81-...
7Eh	7E-7F-78-79-...-7D	7E-7F-70-71-...-7D	7E-7F-WAIT-WAIT-80-81-...
7Fh	7F-78-79-7A-...-7E	7F-70-71-72-...-7E	7F-WAIT-WAIT-WAIT-80-81

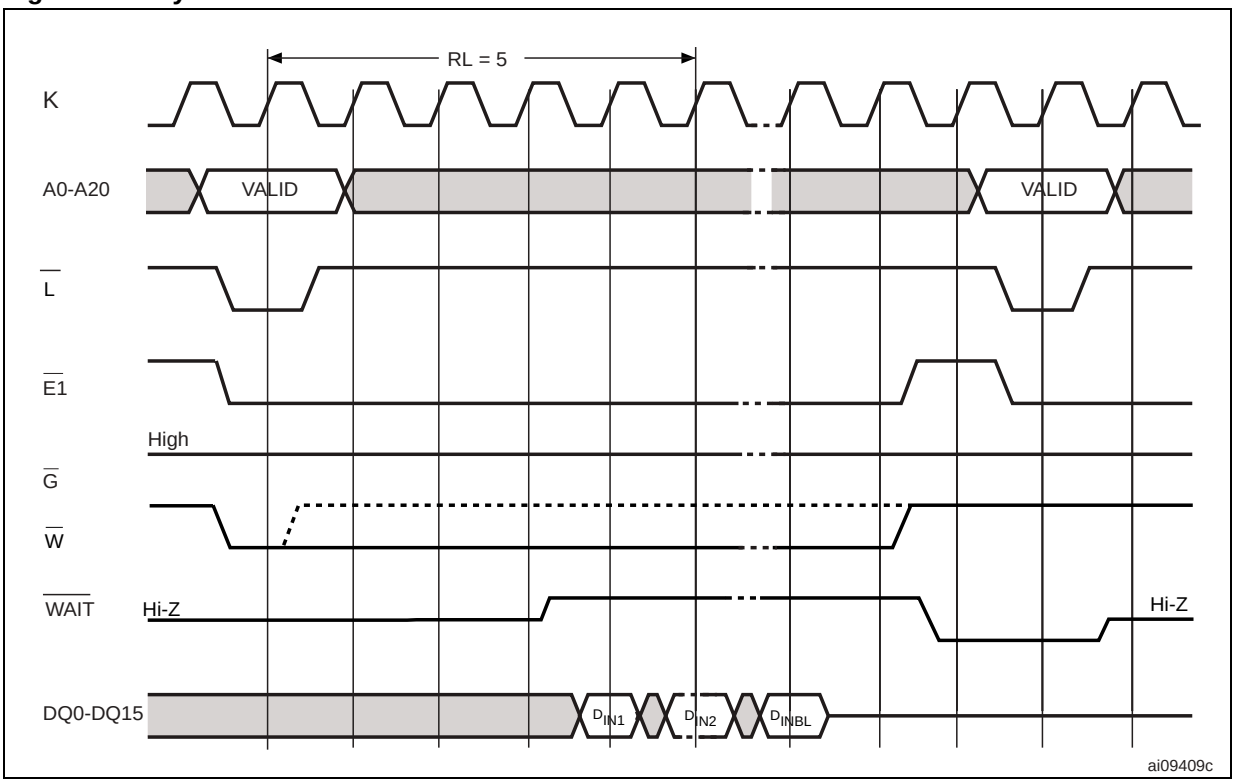
1. Read address is in hexadecimal format.

Figure 4. Synchronous Burst Read



1. The above diagram is an example of Synchronous Burst Read waveforms for a Latency of 5 clock cycles.

Figure 5. Synchronous Burst Write



1. The above diagram is an example of Synchronous Burst Write waveforms for a Latency of 5 clock cycles.

Figure 6. Burst Length Definition

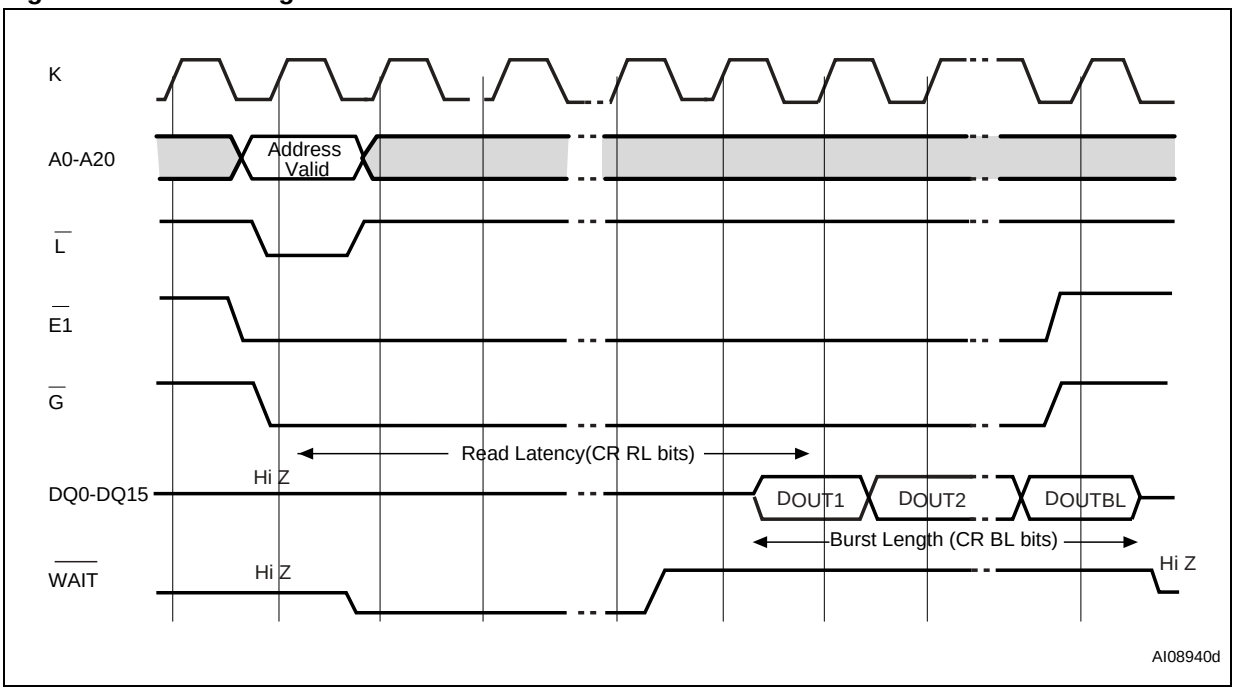


Figure 7. Latency definition

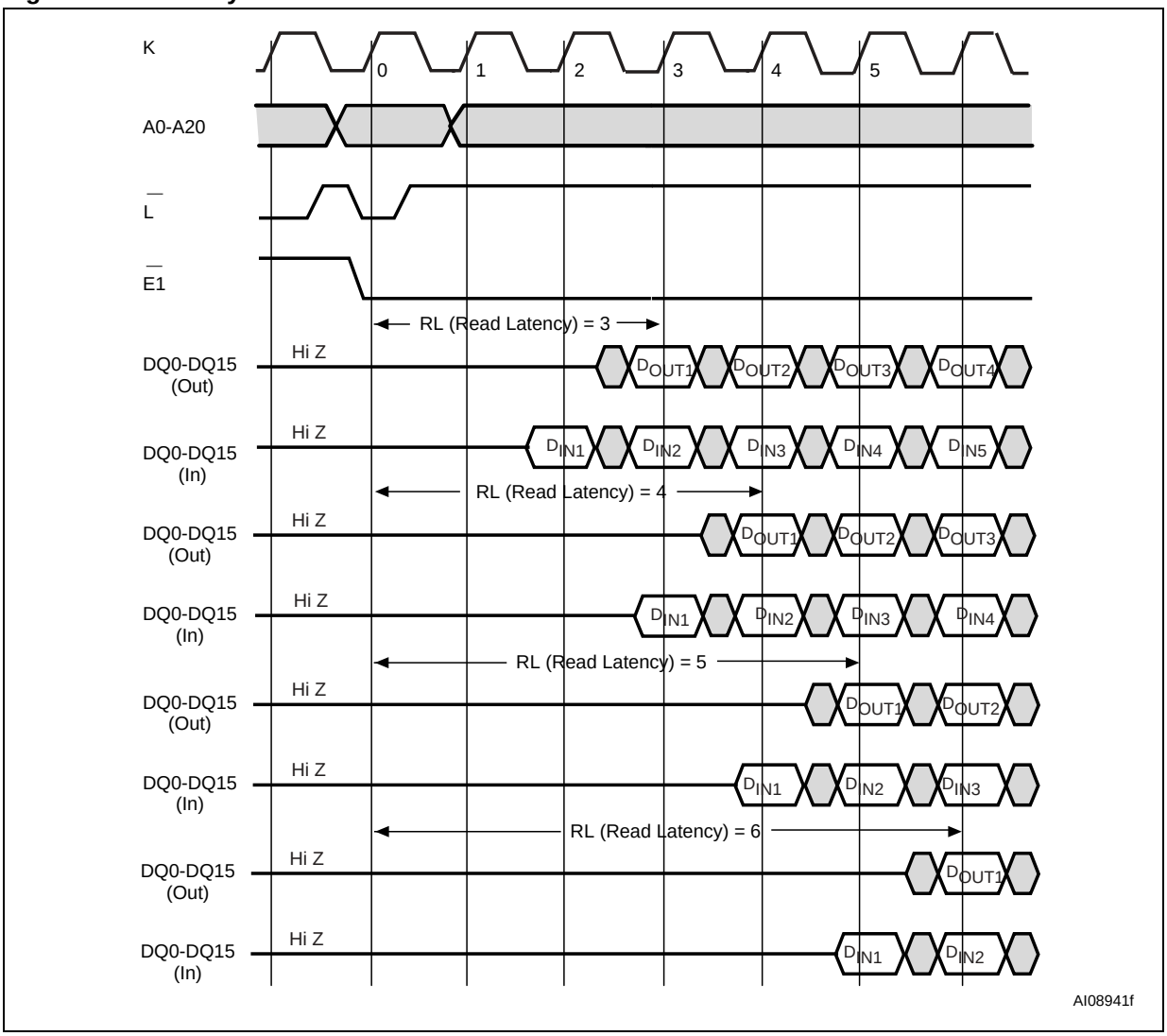
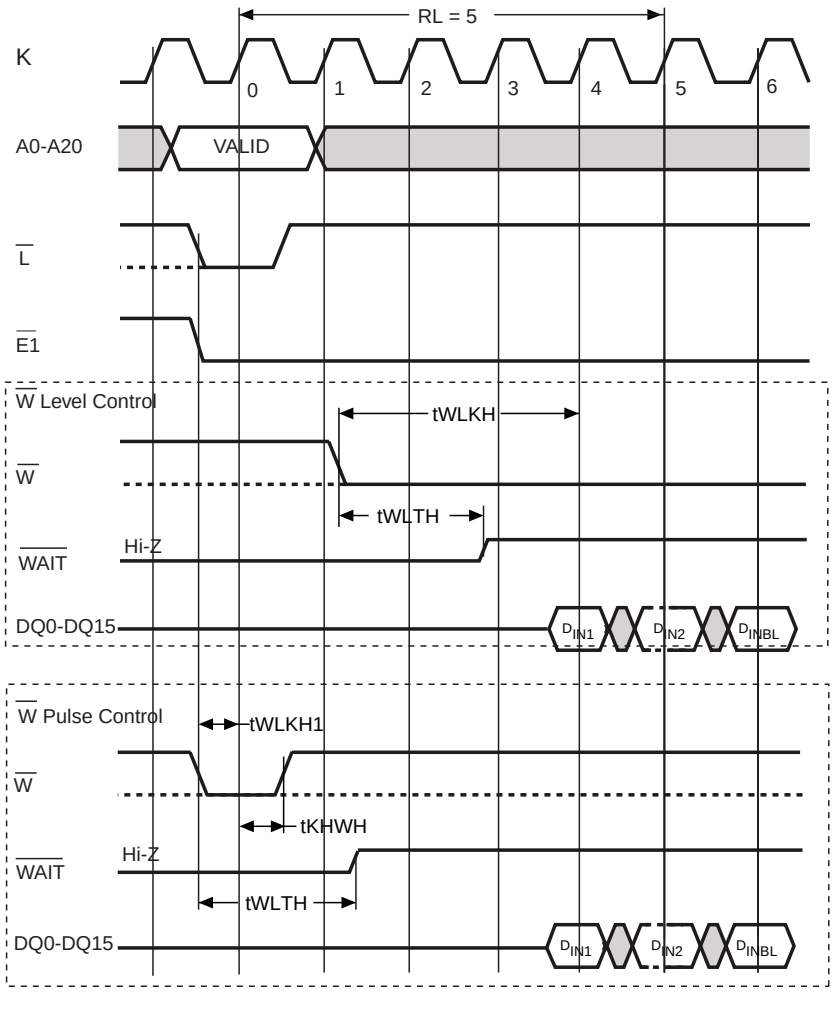


Figure 8. Write Enable control modes



1. The above diagram is given as an example for a Latency of 5 clock cycles.

4 Maximum rating

Stressing the device above the rating listed in the Absolute Maximum Ratings table may cause permanent damage to the device. Exposure to Absolute Maximum Rating conditions for extended periods may affect device reliability. These are stress ratings only and operation of the device at these or any other conditions above those indicated in the Operating sections of this specification is not implied. Refer also to the STMicroelectronics SURE Program and other relevant quality documents.

Table 8. Absolute maximum ratings

Symbol	Parameter	Min	Max	Unit
I_O	Output Current	– 50	50	mA
T_A	Ambient Operating Temperature	– 30	85	°C
T_{STG}	Storage Temperature	– 55	125	°C
V_{CC}	Core Supply Voltage	– 0.5	3.6	V
V_{IO}	Input or Output Voltage	– 0.5	3.6	V

5 DC and AC parameters

This section summarizes the operating measurement conditions, and the DC and AC characteristics of the device. The parameters in the DC and AC characteristics Tables that follow, are derived from tests performed under the Measurement Conditions summarized in *Table 9: Operating and AC measurement conditions*. Designers should check that the operating conditions in their circuit match the operating conditions when relying on the quoted parameters.

Table 9. Operating and AC measurement conditions

Parameter	M69KB048BD, M69AB048BD		Unit
	Min	Max	
V_{CC} Supply Voltage(1)	– 1.7	1.95	V
Ambient Operating Temperature	– 30	85	°C
Load Capacitance (C_L)	50		pF
Output Circuit Resistance (R_1)	50		Ω
Input Pulse Voltages(1)	0	V_{CC}	V
Input and Output Timing Ref. Voltages(1)	V_{CC}/ 2		V
Output Transition Timing Ref. Voltages(1)	$0.3V_{CC}$	$0.7V_{CC}$	V
Junction Temperature (T_J)	–30	90	°C

1. All voltages are referenced to V_{SS}.

Figure 9. AC measurement I/O waveform

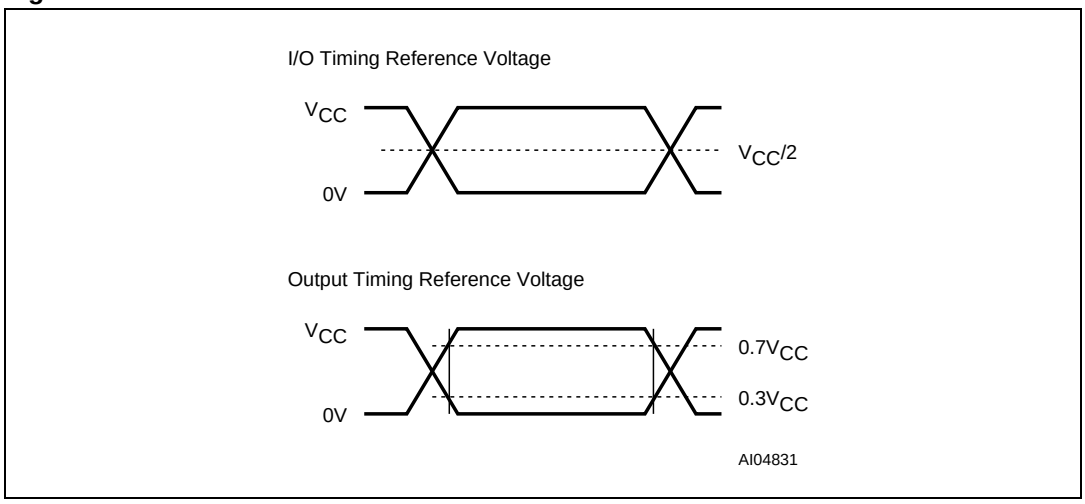


Figure 10. AC measurement load circuit

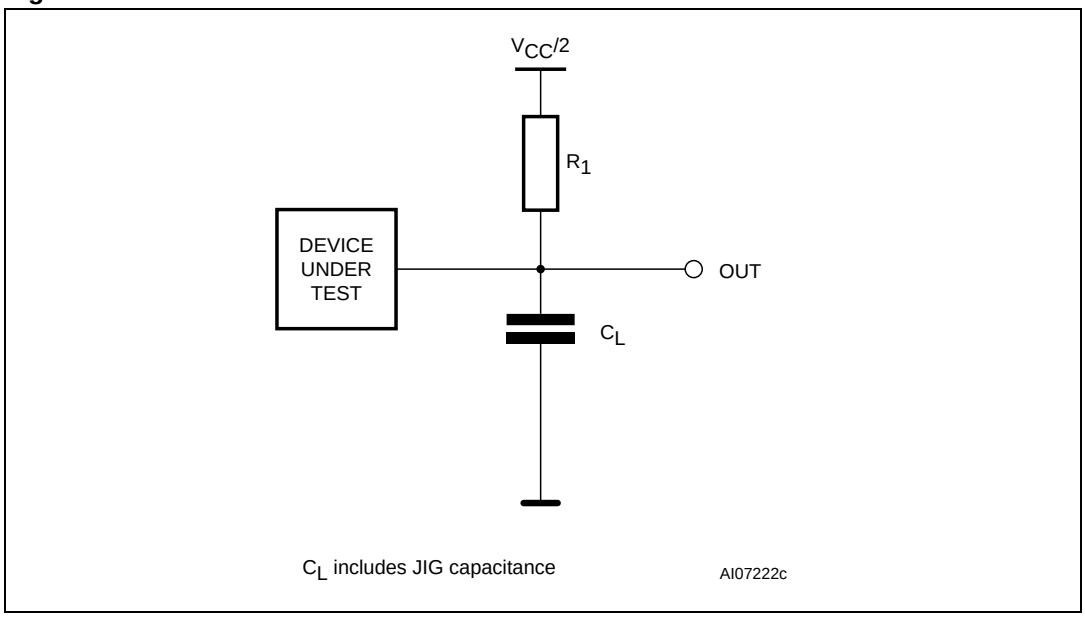


Table 10. Capacitance

Symbol	Parameter(1)	Test Condition	Min	Max	Unit
C_{IN1}	Address Input Capacitance	V_{IN} = 0V		5	pF
C_{IN2}	Control Input Capacitance	V_{IN} = 0V		5	pF
C_{IO}	Data Input/Output Capacitance	V_{IO} = 0V		8	pF

1. T_A = 25°C, f = 1 MHz

Table 11. DC characteristics

Symbol	Parameter	Test Condition(1)		Min	Max	Unit
I_{CC}	V_{CC} Active Current	V_{CC} = 1.95V, V_{IN} = V_{IH} or V_{IL}, $\overline{E1}$ = V_{IL} and E2 = V_{IH}, I_{OUT} = 0mA	t_{AVAX} = minimum		30	mA
			t_{AVAX} = 1 µs		3	mA
I_{CCP}	V_{CC} Active Current (Page Read)	V_{CC} = 1.95V, V_{IN} = V_{IH} or V_{IL}, $\overline{E1}$ = V_{IL} and E2 = V_{IH}, I_{OUT} = 0mA, t_{AVAX2} = min.			10	mA
I_{CCB}	V_{CC} Active Current (Burst Read)	V_{CC} = 1.95V, V_{IN} = V_{IH} or V_{IL}, $\overline{E1}$ = V_{IL} and E2 = V_{IH}, BL = Continuous, t_K(2)=t_{KHKH} min I_{OUT} = 0mA			25	mA

Table 11. DC characteristics (continued)

Symbol	Parameter	Test Condition(1)		Min	Max	Unit
IPAR	V_{CC} Power Down Current in Deep Power-Down mode (ICCPD)	V_{CC} = 1.95V, V_{IN} = V_{IH} or V_{IL}, E2 ≤ 0.2V			10	µA
	V_{CC} Power Down Current in 4Mb PAR mode (ICCP4)				45	µA
	V_{CC} Power Down Current in 8Mb PAR mode (ICCP8)				55	µA
I_{SB}	V_{CC} Standby Current	V_{CC} = 1.95V, V_{IN} (including K) = V_{IH} or V_{IL}, $\overline{E1}$= E2= V_{IH}			1.5	mA
		V_{CC} = 1.95V, V_{IN} (including K) ≤ 0.2V or V_{IN} (including K)≥ V_{CC} – 0.2V $\overline{E1}$= E2 ≥ V_{CC}–0.2V			100	µA
		V_{CC} = 1.95V, t_K=t_{KHKH} min(2) V_{IN} ≤ 0.2V or V_{IN} ≥ V_{CC} – 0.2V $\overline{E1}$=E2 ≥ V_{CC} – 0.2V	RL = 6		500	µA
			RL = 3,4,5		220	
I_{LI}	Input Leakage Current	0V ≤ V_{IN} ≤ V_{CC}		– 1.0	1.0	µA
I_{LO}	Output Leakage Current	0V ≤ V_{OUT} ≤ V_{CC} (Output Disabled)		– 1.0	1.0	µA
V_{IH}(3)(4)	Input High Voltage			0.8V_{CC}	V_{CC}– 0.2V	V
V_{IL}(3)(4)	Input Low Voltage			– 0.3	0.2V_{CC}	V
V_{OH}(3)	Output High Voltage	V_{CC} = 1.7V, I_{OH} = –0.5mA		1.4		V
V_{OL}(3)	Output Low Voltage	I_{OL} = 1mA			0.4	V

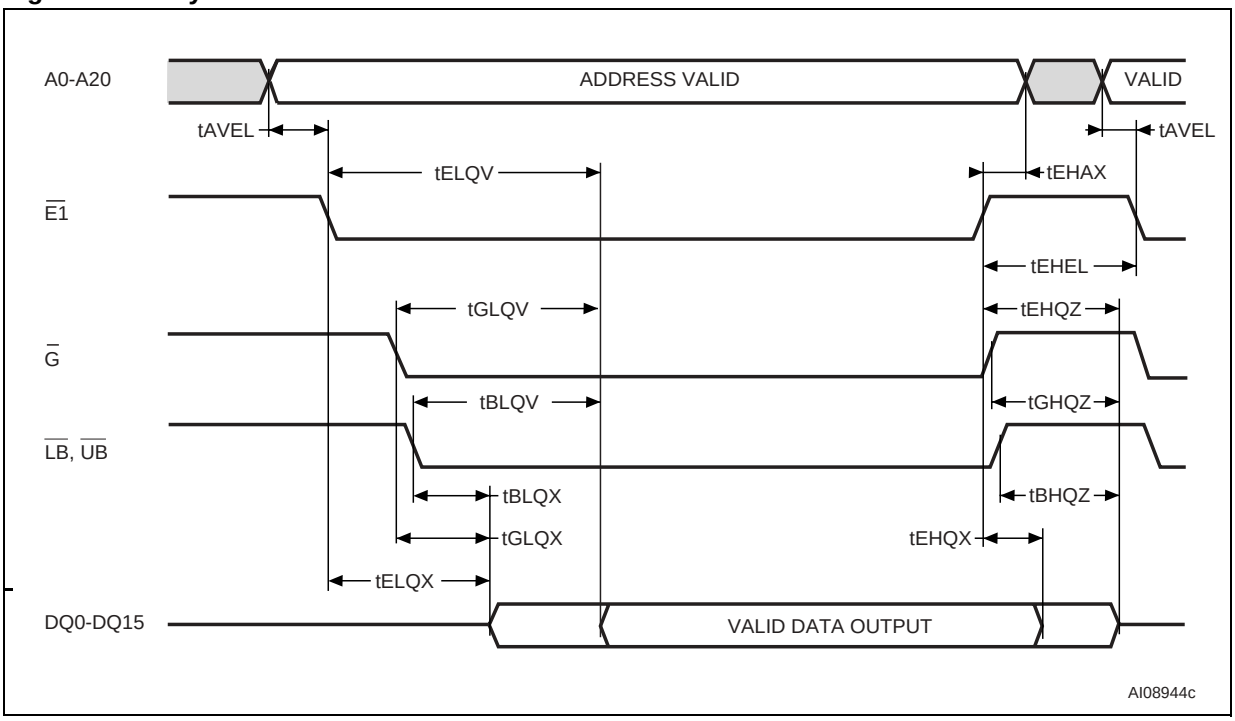
1. DC Characteristics are measured after power-up.
2. t_{KHKH} is the clock period.
3. All voltages are referenced to V_{SS}.
4. During voltage transitions, input may overshoot to V_{CC} + 1.0V and undershoot to V_{SS} – 1.0V for a period of up to 5ns.

Table 12. Asynchronous Read AC characteristics (Page mode)

Symbol	Alt.	Parameter	M69KB048BD, M69AB048BD		Unit
			Min	Max	
t_{AVAX}(1)(2)	t_{RC}	Read Cycle Time	70	1000	ns
t_{ELAX}			70	1000	ns
t_{ELQV}(3)	t_{CE}	Chip Enable Low to Output Valid		70	ns
t_{GLQV}(3)	t_{OE}	Output Enable Low to Output Valid		40	ns
t_{AVQV}(3)(4)	t_{AA}	Address Valid to Output Valid		70	ns
t_{BLQV}(3)	t_{BA}	Upper/Lower Byte Enable Low to Output Valid		30	ns
t_{AVQV2}(3)(5)	t_{PAA}	Page Address Access Time		20	ns
t_{AVAX2}, t_{AVEH} (1)(5)(6)	t_{PRC}	Page Read Cycle Time	20	1000	ns
t_{AXQX}(3)	t_{OH}	Data hold from address change	5		ns
t_{EHQX}(3)		Data hold from Chip Enable High	5		ns
t_{GHQX}(3)		Data hold from Chip Enable High	5		ns
t_{ELQX}(7)	t_{CLZ}	Chip Enable Low to Output Transition	5		ns
t_{GLQX}(7)	t_{OLZ}	Output Enable Low to Output Transition	10		ns
t_{BLQX}(7)	t_{BLZ}	Upper/Lower Byte Enable Low to Output Transition	0		ns
t_{EHQZ}(7)	t_{CHZ}	Chip Enable High to Output Hi-Z		12	ns
t_{GHQZ}(7)	t_{OHZ}	Output Enable High to Output Hi-Z		12	ns
t_{BHQZ}(7)	t_{BHZ}	Upper/Lower Byte Enable High to Output Hi-Z		12	ns
t_{AVEL}	t_{ASC}	Address Valid to Chip Enable Low	–5		ns
t_{AVGL}	t_{ASO}	Address Valid to Output Enable Low	0		
t_{AXAV}(4)(8)	t_{AX}	Address Invalid Time		10	ns
t_{EHAX}(9)	t_{CHAH}	Chip Enable High to Address Invalid	– 5		ns
t_{GHAX}	t_{OHAH}	Output Enable High to Address Invalid	– 5		ns
t_{WHGL}(10)	t_{WHOL}	Write Enable High to Output Enable Low	10	1000	ns
t_{EHEL}	t_{CP}	Chip Enable High Pulse Width	15		ns

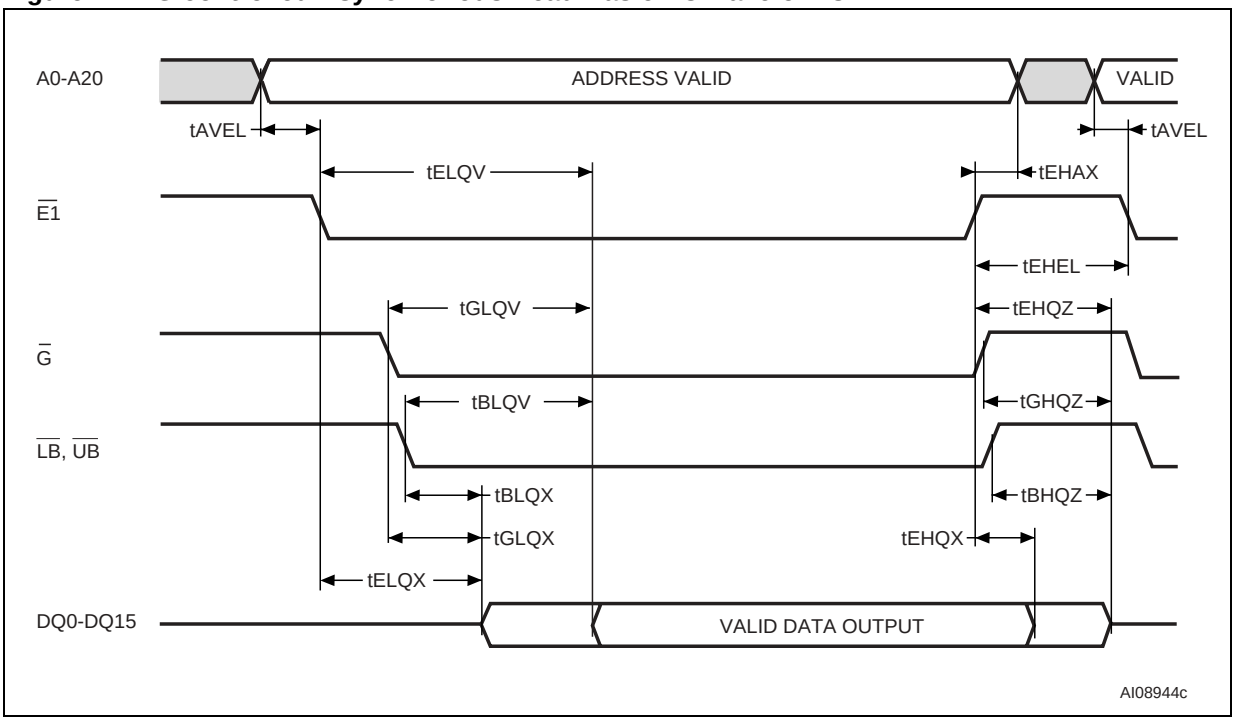
1. Maximum value is applicable if $\overline{E1}$ is kept Low without change of address input of A3 to A20.
2. Address should not be changed within minimum Read Cycle Time (t_{AVAX}).
3. See *Figure 14*.
4. Applicable to A3 to A20 when $\overline{E1}$ is kept Low.
5. Applicable only to A0, A1 and A2 when $\overline{E1}$ is kept Low for the page address access.
6. In case Page Read Cycle is continued with keeping $\overline{E1}$ stays Low, $\overline{E1}$ must be brought to High within 4µs. In other words, Page Read Cycle must be closed within 4µs.
7. The output Load capacitance is 5pF without any other load.
8. Applicable when at least two of address inputs are switched from previous state.
9. Minimum Read Cycle TIme, t_{AVAX}, and minimum Page Read Cycle Time, t_{AVAX2} must be satisfied.
10. If t_{WHGL} is shorter than the minimum value specified in *Table 12*, the address access time, t_{AVQV} of the following Read operations may become longer by $t_{WHGL(min)} - t_{WHGL}$.

Figure 11. Asynchronous Read Basic AC waveforms



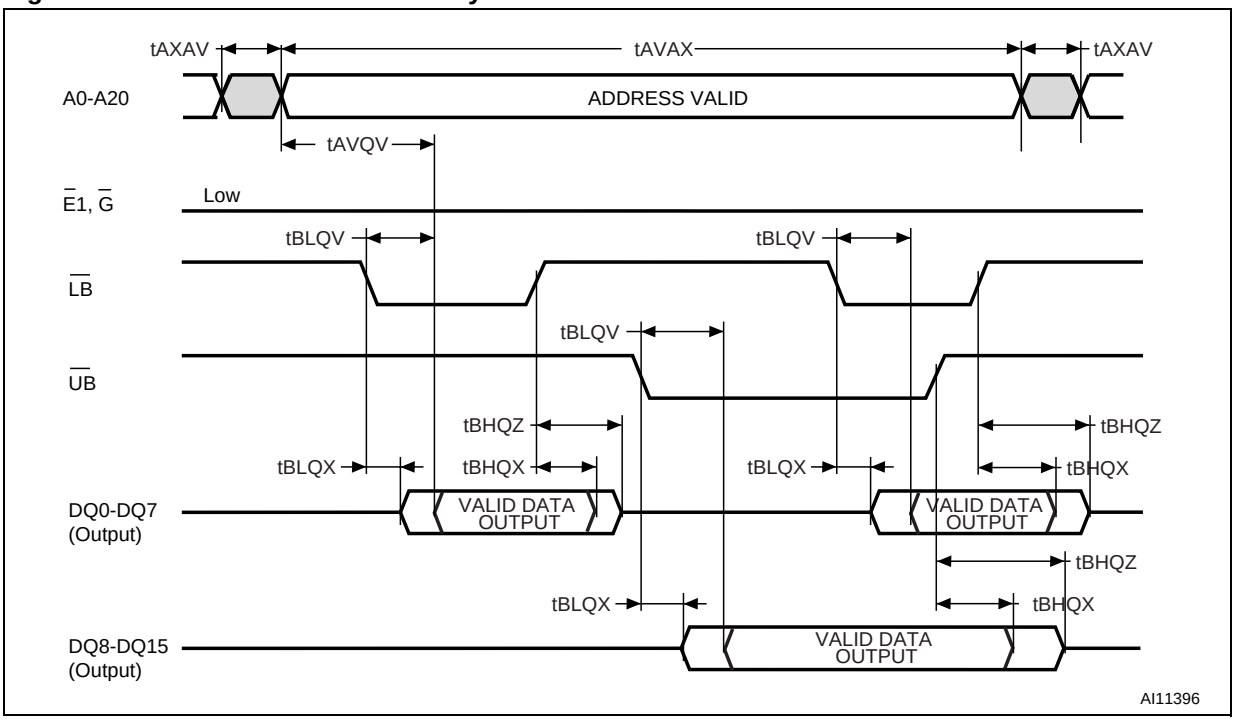
1. E2 High, $\overline{W}$ High, $\overline{L}$ Low.

Figure 12. $\overline{G}$ controlled Asynchronous Read Basic AC waveforms



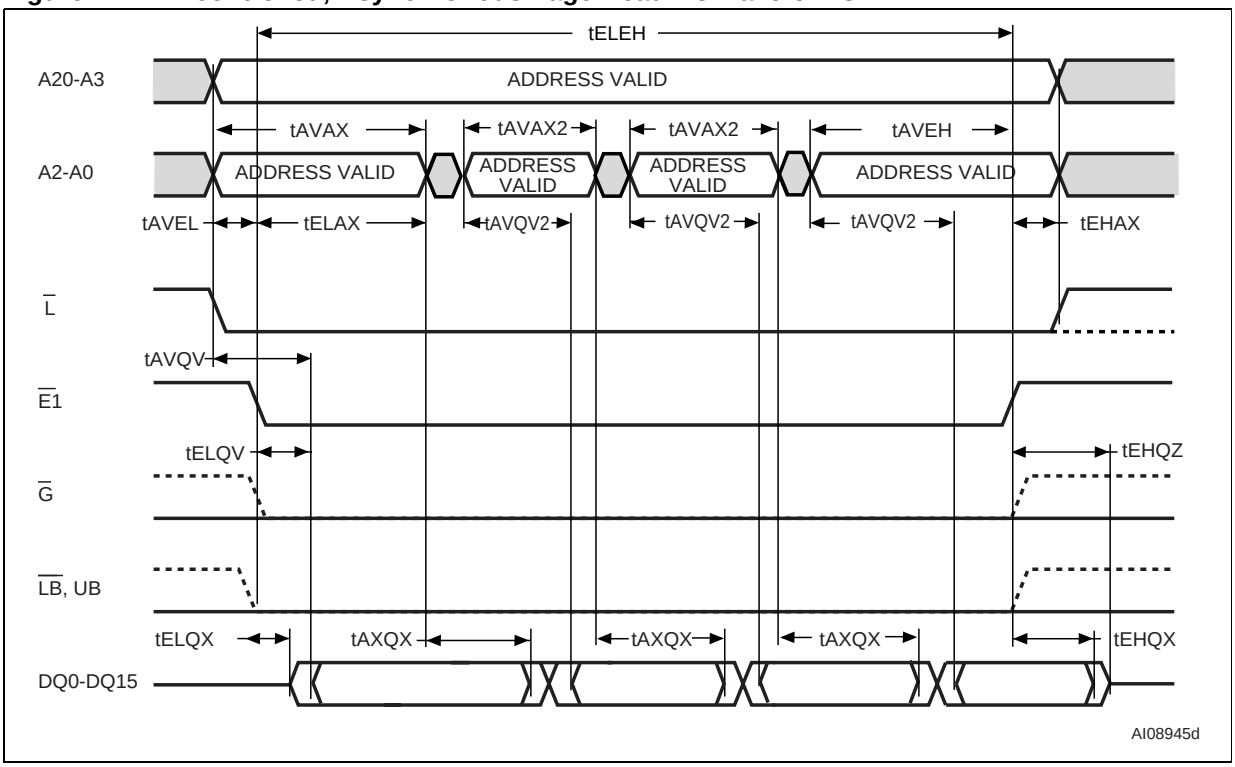
1. E2 High, $\overline{W}$ High, $\overline{L}$ Low.

Figure 13. $\overline{LB}$ and $\overline{UB}$ controlled Asynchronous Read basic AC waveforms



1. E2 High, $\overline{W}$ High, $\overline{L}$ Low.

Figure 14. $\overline{E1}$ controlled, Asynchronous Page Read AC waveforms



1. Write Enable ($\overline{W}$) = High, E2 = High.

Table 13. Asynchronous Write mode AC characteristics

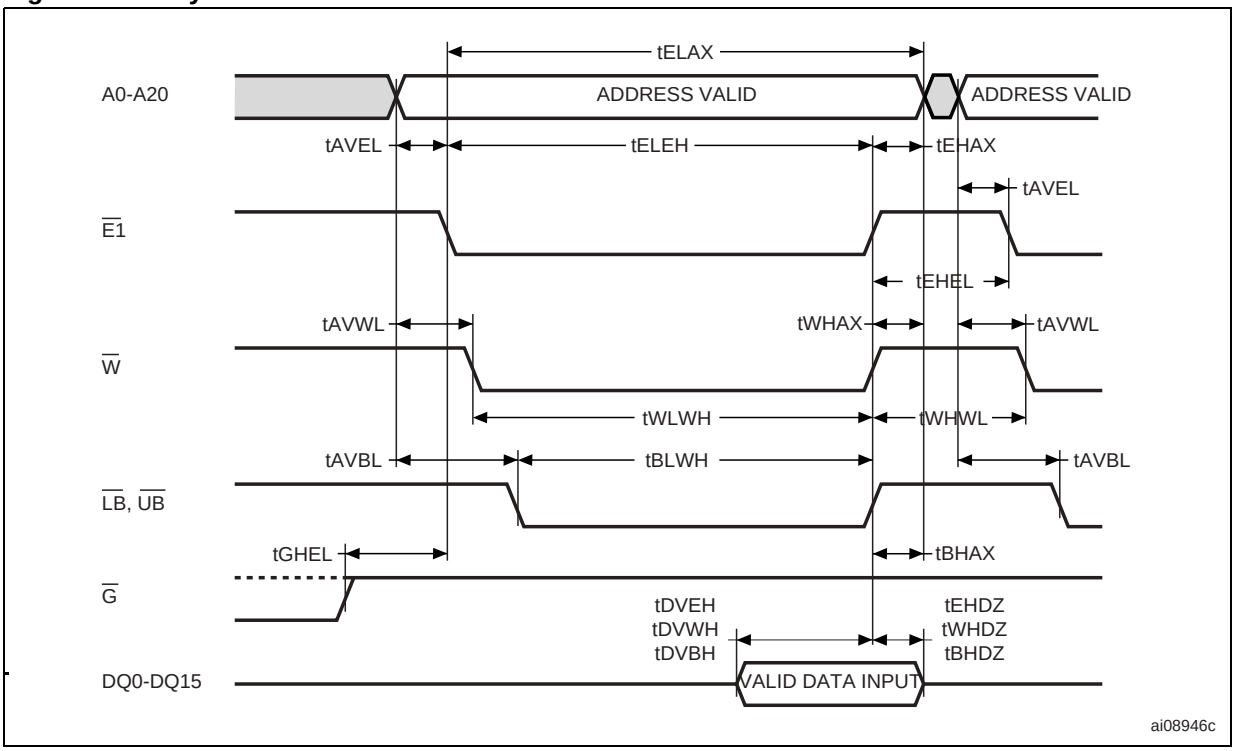
Symbol	Alt.	Parameter	M69KB048BD, M69AB048BD		Unit
			Min	Max	
t_{AVAX}(1)(2)	t_{WC}	Write Cycle Time	70	1000	ns
t_{ELAX}			70	1000	ns
t_{AVBL}(3)	t_{AS}	Address Valid to $\overline{LB}$, $\overline{UB}$ Low	0		ns
t_{AVEL}(3)		Address Valid to Chip Enable Low	0		ns
t_{AVWL}(3)		Address Valid to Write Enable Low	0		ns
t_{ELEH}(3)	t_{CW}	Chip Enable Low to Chip Enable High	45		ns
t_{LLLH}(3)	t_{VPL}	Latch Enable Low Pulse width	10		ns
t_{LHLL}(3)	t_{VPH}	Latch Enable High Pulse width	10		ns
t_{BHEL}, t_{BHLL} (4)	t_{BS}	$\overline{LB}$ / $\overline{UB}$ Byte Mask Setup Time	–5		ns
t_{WHBL}(5)	t_{BH}	$\overline{LB}$ / $\overline{UB}$ Byte Mask Hold Time	–5		ns
t_{WLBH}(3)	t_{WP}	Write Enable Low to $\overline{LB}$, $\overline{UB}$ High	45		ns
t_{WLWH}		Write Enable Low to Write Enable High	45		ns
t_{BLWH}(3)	t_{BW}	$\overline{LB}$, $\overline{UB}$ Low to Write Enable High	45		ns
t_{BLBH}(3)		$\overline{LB}$, $\overline{UB}$ Low to $\overline{LB}$, $\overline{UB}$ High	45		ns
t_{EHAX}(6)	t_{WRC}	Chip Enable High to Address Transition	15		ns
t_{WHAX}(6)	t_{WR}	Write Enable High to Address Transition	0		ns
t_{BHAX}(6)	t_{BR}	$\overline{LB}$, $\overline{UB}$ High to Address Transition	15	1000	ns
t_{DVEH}	t_{DS}	Input Valid to Chip Enable High	15		ns
t_{DVWH}		Input Valid to Write Enable High	15		ns
t_{DVBH}		Input Valid to $\overline{LB}$, $\overline{UB}$ High	15		ns
t_{EHDZ}	t_{DH}	Chip Enable High to Input High-Z	0		ns
t_{WHDZ}		Write Enable High to Input High-Z	0		ns
t_{BHDZ}		$\overline{LB}$, $\overline{UB}$ High to Input High-Z	0		ns
t_{GHEL}(7)	t_{OHCL}	Output Enable High to Chip Enable Low	–5		ns

Table 13. Asynchronous Write mode AC characteristics (continued)

Symbol	Alt.	Parameter	M69KB048BD, M69AB048BD		Unit
			Min	Max	
t_{GHAV}(8)	t_{OES}	Output Enable High to Address Valid	0		ns
t_{BLBH2}	t_{BWO}	$\overline{LB}$, $\overline{UB}$ Low to $\overline{LB}$, $\overline{UB}$ High for Page Access	30		ns
t_{EHEL}	t_{CP}	Chip Enable High to Chip Enable Low	10		ns
t_{WHWL}	t_{WHP}	Write Enable High Pulse Width	10	1000	ns

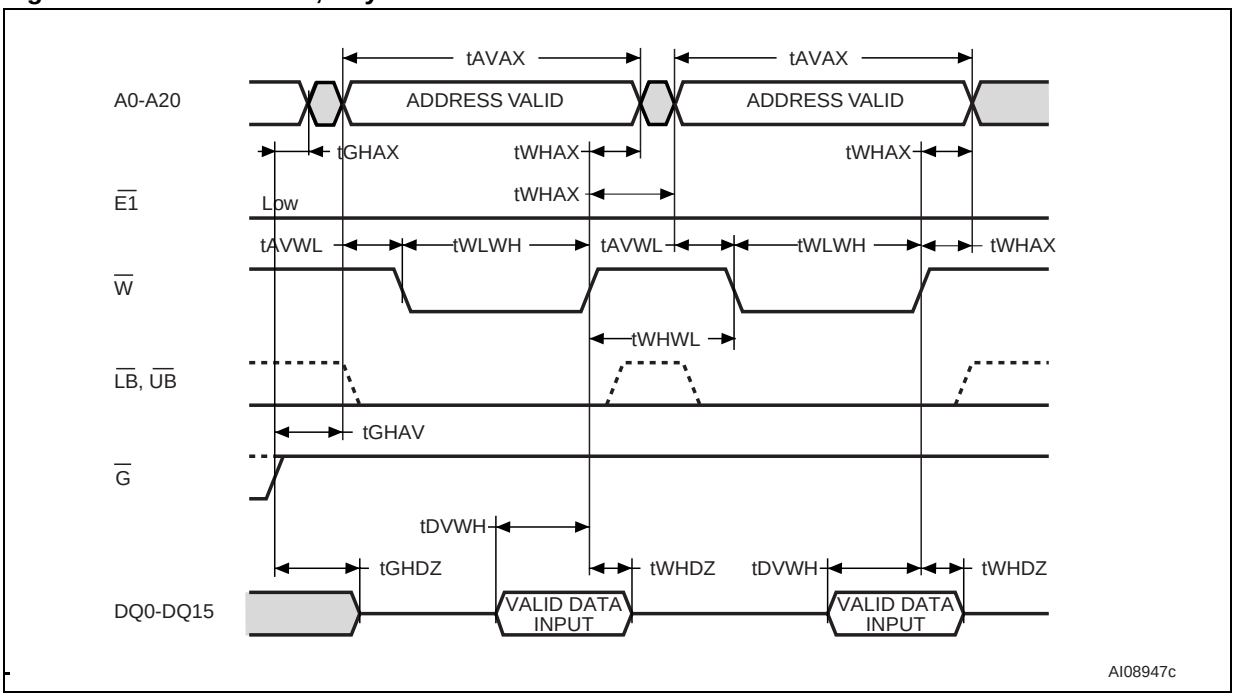
1. Maximum value is applicable if $\overline{E1}$ is kept Low without any address change. If needed by system operation, please contact your local ST representative for relaxation of the 1000ns limitation.
2. Minimum value must be equal to or greater than the sum of write pulse (t_{ELEH}, t_{WLBH} or t_{BLBH}) and write recovery time (t_{EHAX}, t_{WHAX} or t_{BHAX}).
3. Write pulse is defined from the falling edge of $\overline{E1}$, $\overline{W}$, or $\overline{LB}$/$\overline{UB}$, whichever occurs last.
4. Applicable only to $\overline{LB}$ and $\overline{UB}$. $\overline{LB}$ and $\overline{UB}$ setup times are defined from the falling edge of Write Enable, $\overline{W}$, or Chip Enable, $\overline{E1}$, whichever occurs last.
5. Applicable only to $\overline{LB}$ and $\overline{UB}$. $\overline{LB}$ and $\overline{UB}$ hold times are defined from the rising edge of Write Enable, $\overline{W}$, or Chip Enable, $\overline{E1}$, whichever occurs first.
6. Write recovery is defined from Write pulse is defined from the rising edge of $\overline{E1}$, $\overline{W}$, or $\overline{LB}$/$\overline{UB}$, whichever occurs first.
7. If $\overline{G}$ is Low after minimum t_{GLEH}, the read cycle is initiated. In other words, $\overline{G}$ must be brought High within 5ns after $\overline{E1}$ is brought Low. Once the read cycle is initiated, new write pulse should be input after minimum Read Cycle Time, $t_{AVAX,}$ is met.
8. If $\overline{G}$ is Low after new address input, the read cycle is initiated. In other words, $\overline{G}$ must be brought High at the same time or before new address valid. Once the read cycle is initiated, new write pulse should be input after minimum Read Cycle Time, $t_{AVAX,}$ is met.

Figure 15. Asynchronous Write AC waveforms



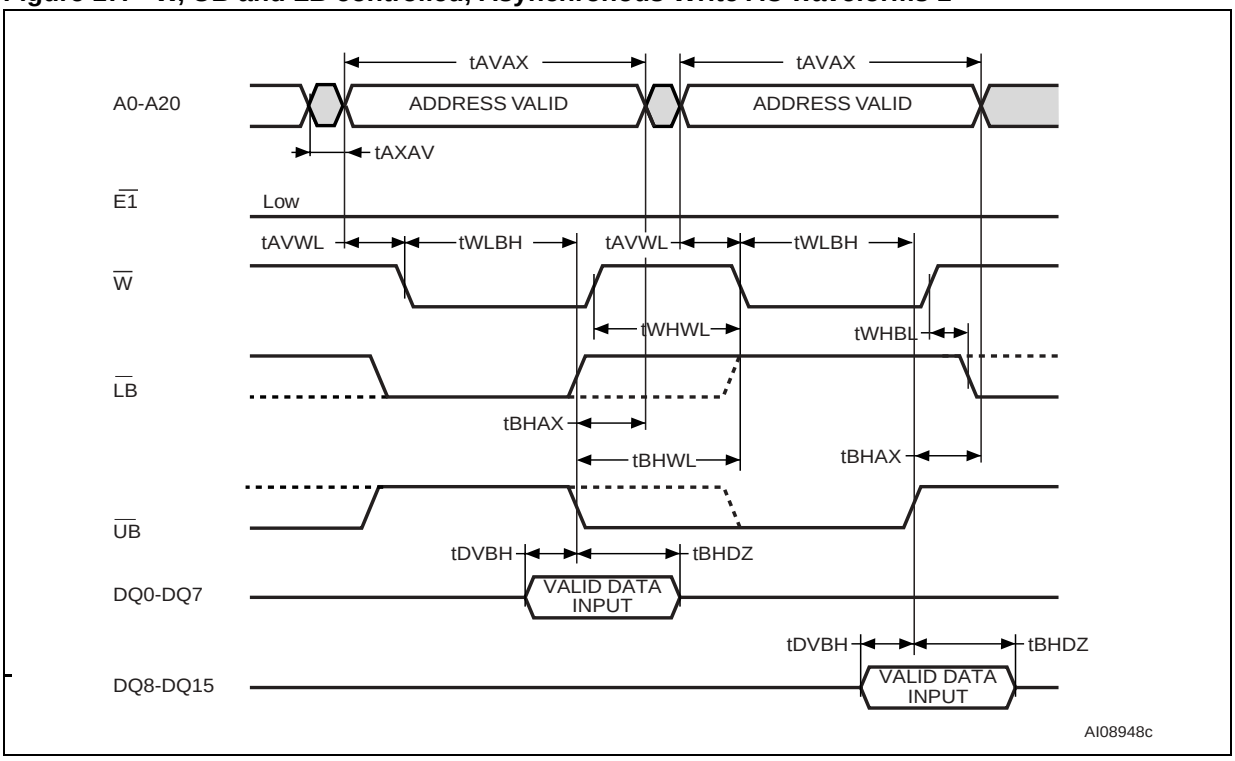
1. E2 = High.

Figure 16. $\overline{W}$ controlled, Asynchronous Write AC waveforms



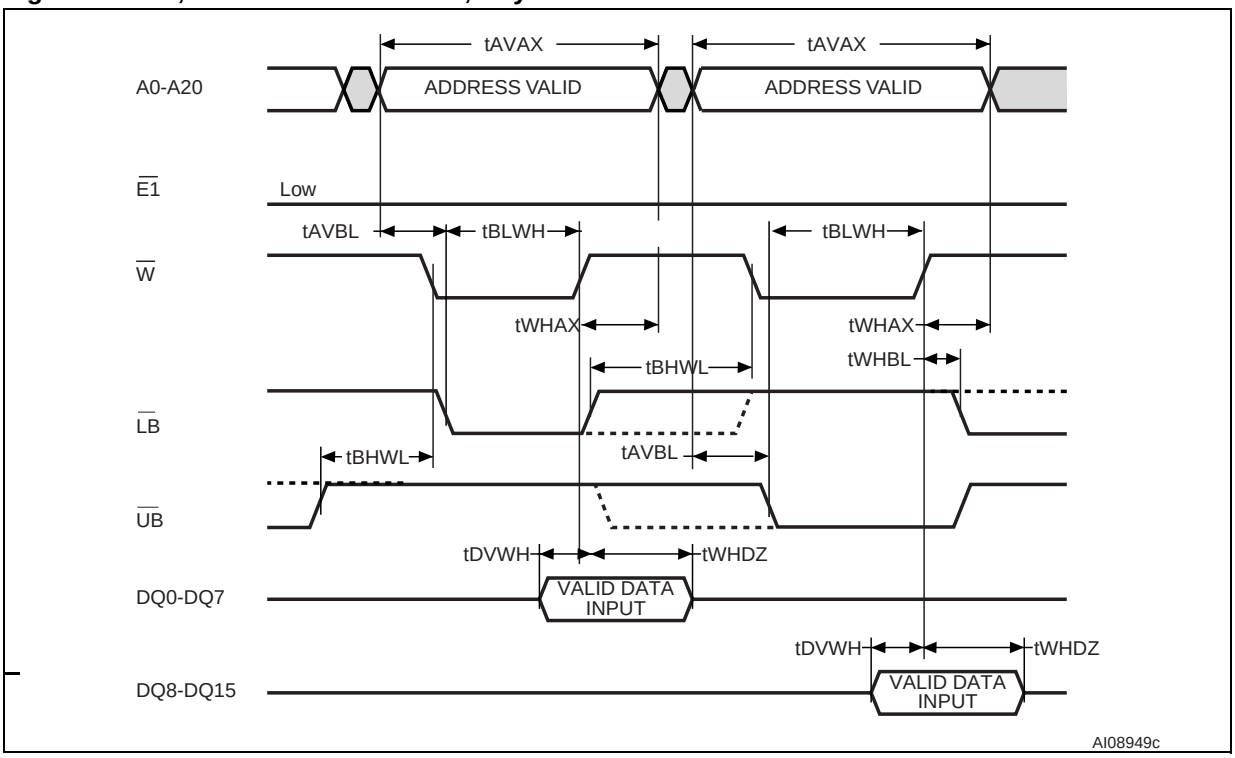
1. E2 = High.

Figure 17. $\overline{W}$, $\overline{UB}$ and $\overline{LB}$ controlled, Asynchronous Write AC waveforms 1



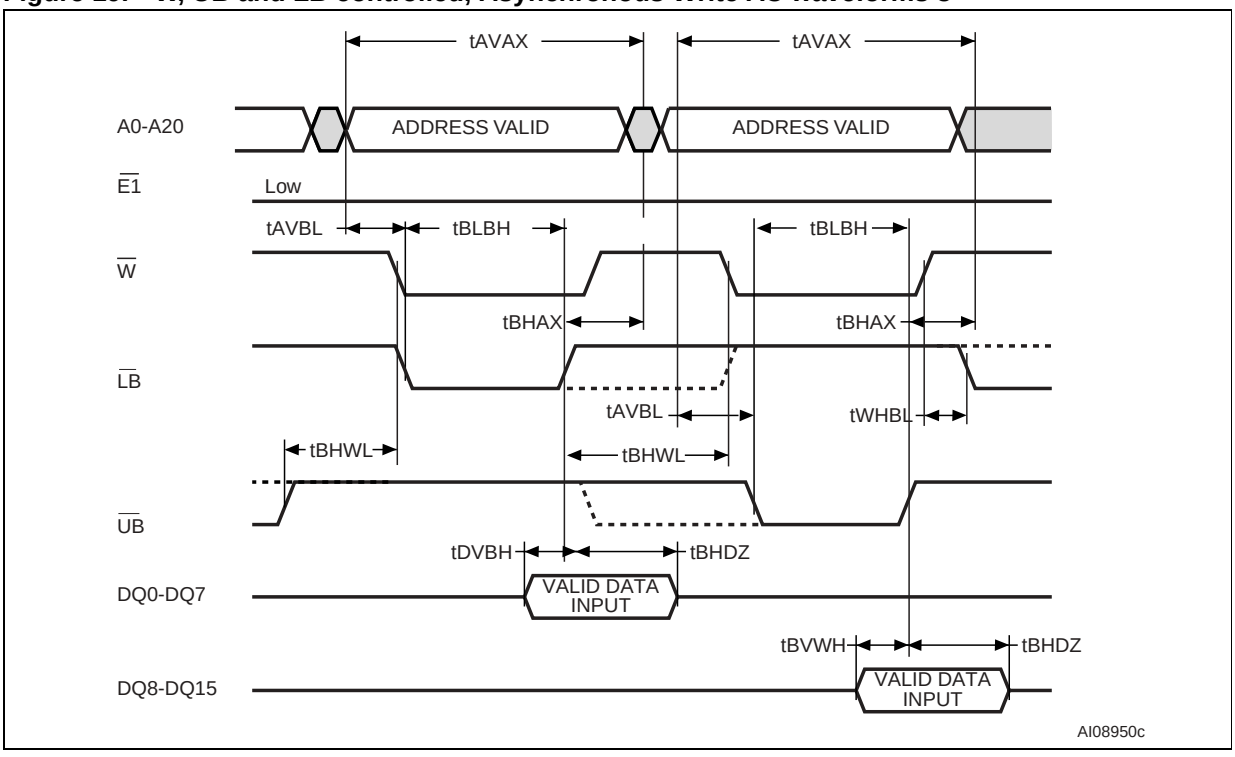
1. E2 = High.

Figure 18. $\overline{W}$, $\overline{UB}$ and $\overline{LB}$ controlled, Asynchronous Write AC waveforms 2



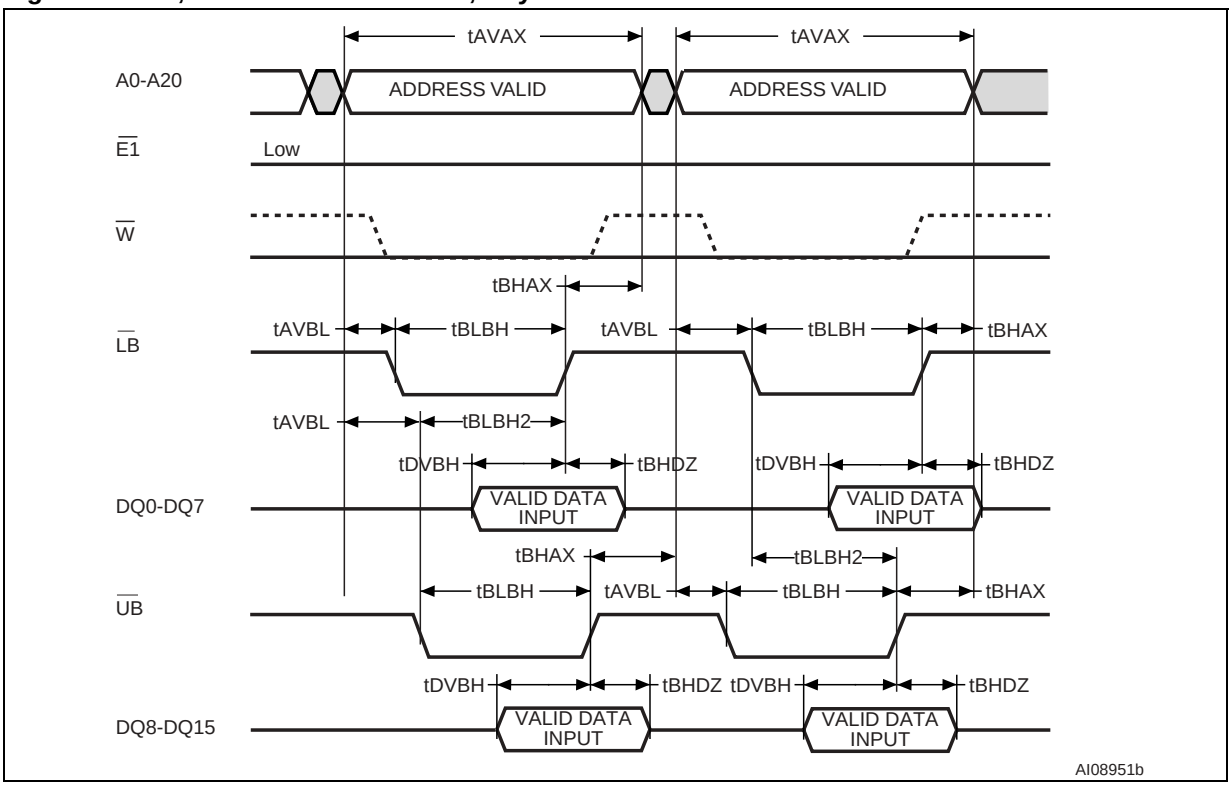
1. E2 = High.

Figure 19. $\overline{W}$, $\overline{UB}$ and $\overline{LB}$ controlled, Asynchronous Write AC waveforms 3



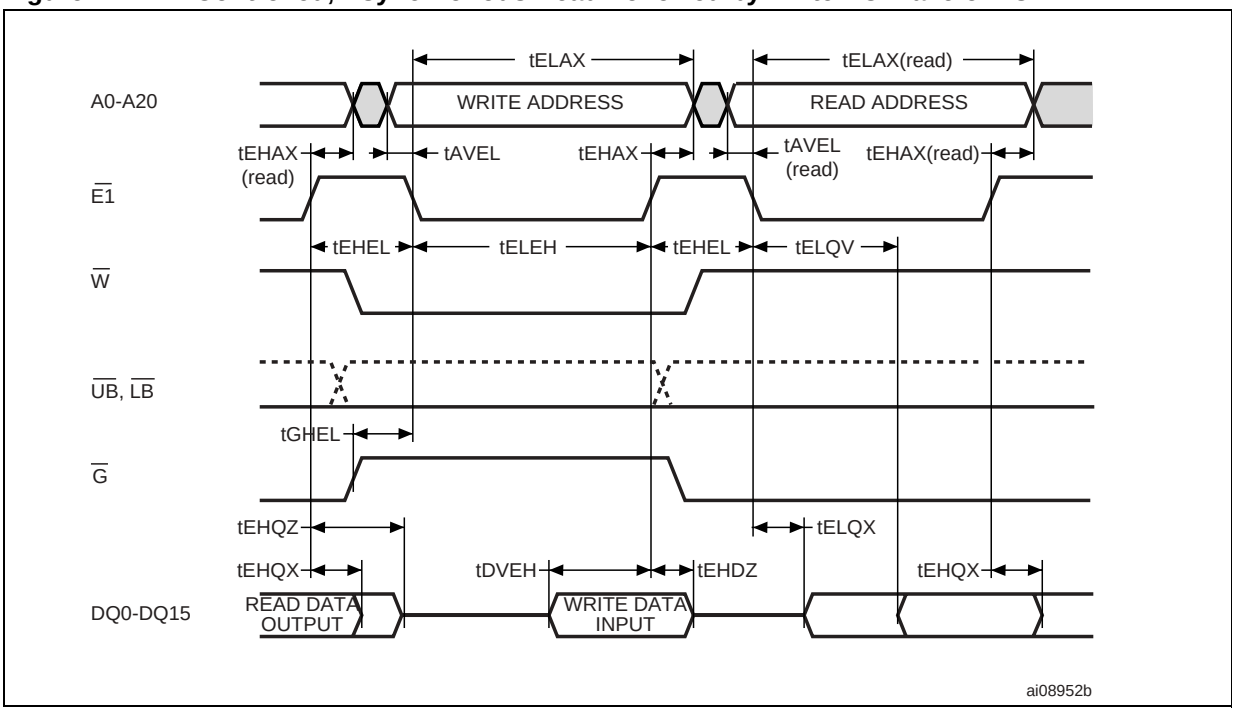
1. E2 = High.

Figure 20. $\overline{W}$, $\overline{UB}$ and $\overline{LB}$ Controlled, Asynchronous Write AC Waveforms 4



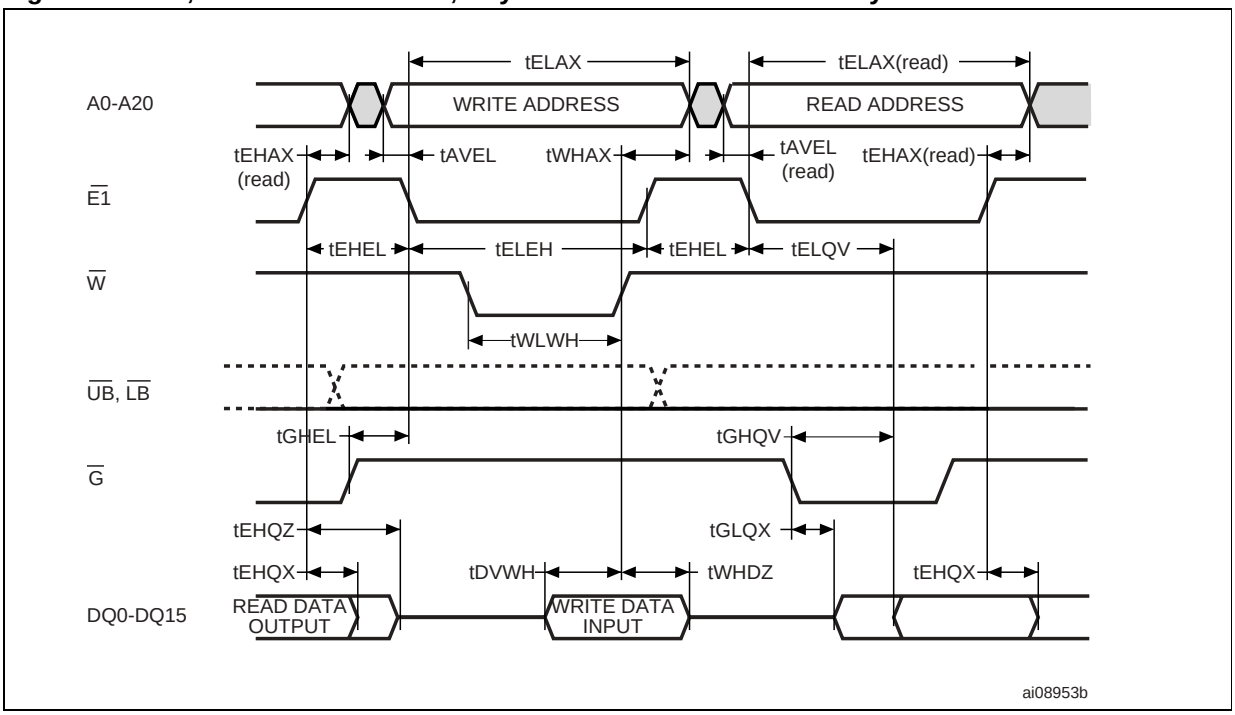
1. E2 = High.

Figure 21. $\overline{E1}$ Controlled, Asynchronous Read Followed by Write AC Waveforms



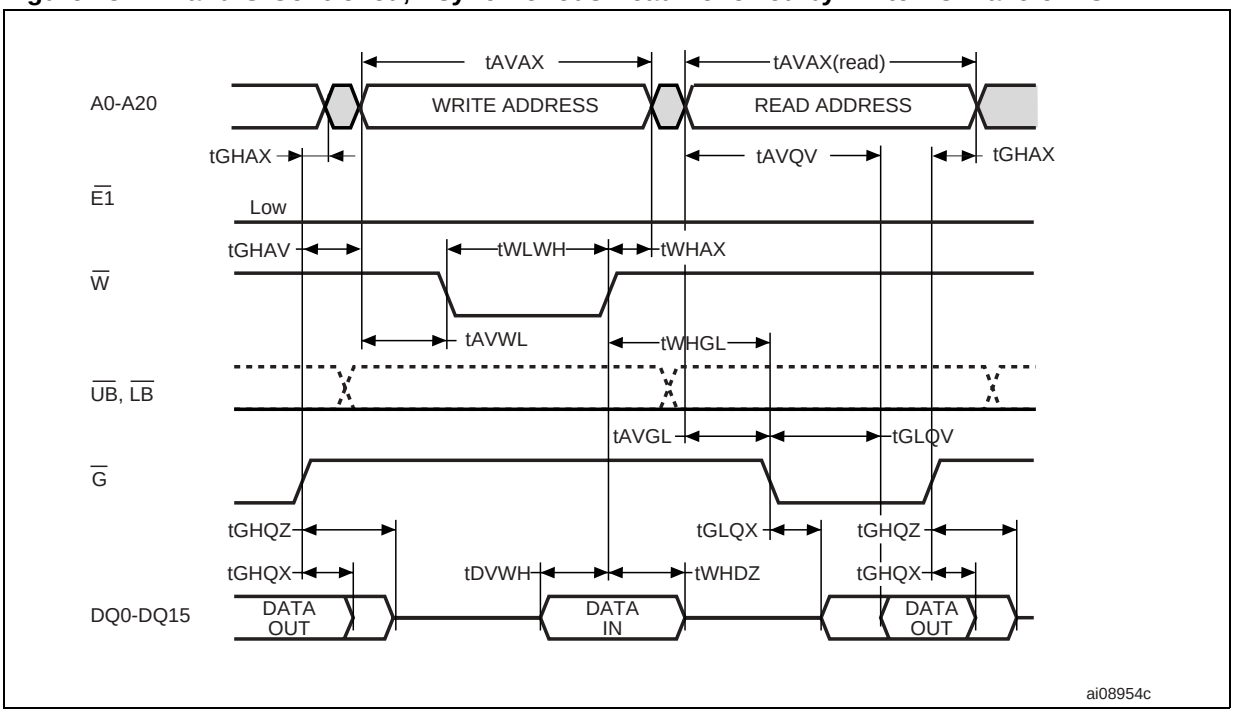
1. Write address is valid from either $\overline{E1}$ or $\overline{W}$ of last falling edge.

Figure 22. $\overline{E1}$, $\overline{W}$ and $\overline{G}$ Controlled, Asynchronous Read Followed by Write AC Waveforms



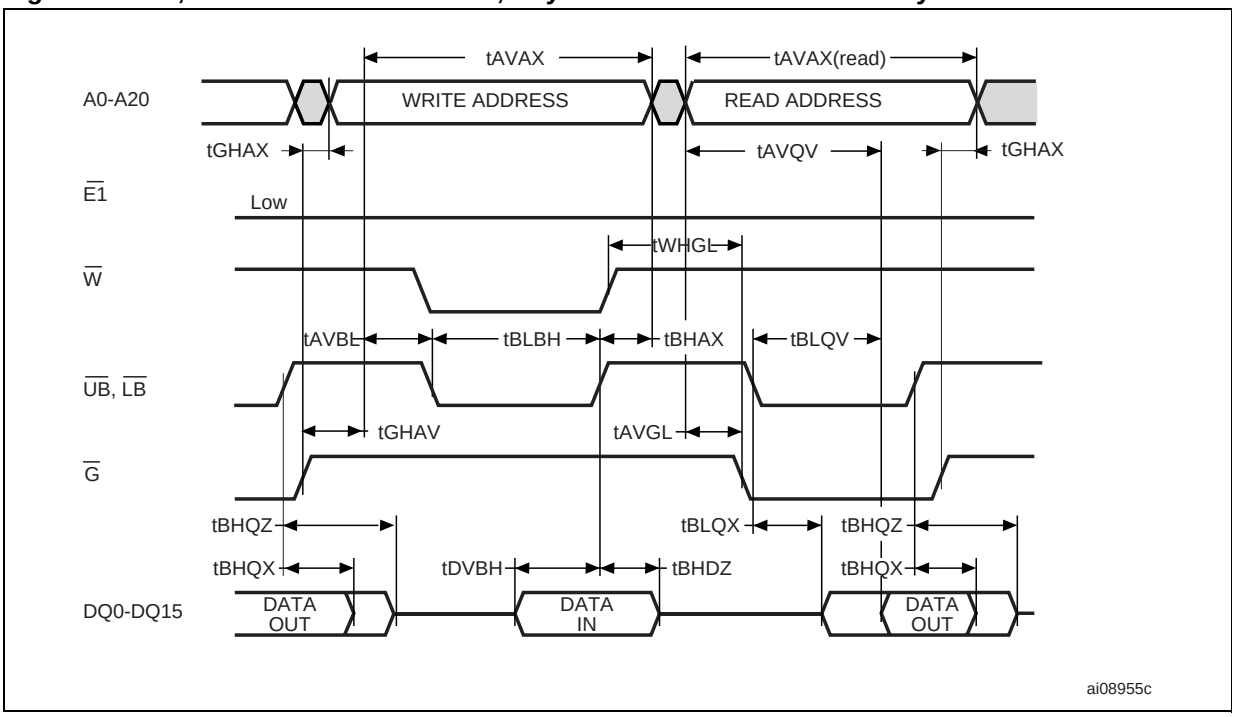
1. $\overline{G}$ can be hold Low during $\overline{E1}$, $\overline{W}$, $\overline{G}$ controlled Write operation.

Figure 23. $\overline{W}$ and $\overline{G}$ Controlled, Asynchronous Read Followed by Write AC Waveforms



1. $\overline{E1}$ can be hold Low for $\overline{W}$ and $\overline{G}$ controlled operations. When $\overline{E1}$ is hold Low, the data outputs are exclusively controlled by $\overline{G}$.

Figure 24. $\overline{W}$, $\overline{G}$ and $\overline{UB}$/$\overline{LB}$ Controlled, Asynchronous Read Followed by Write AC Waveforms



1. $\overline{E1}$ can be hold Low for $\overline{W}$ and $\overline{G}$ controlled operation. When $\overline{E1}$ is hold Low, data outputs are exclusively controlled by $\overline{G}$.

Table 14. Synchronous mode - Clock AC characteristics

Symbol	Alt	Description		Min	Max	Unit
t_{KHKH}	t_{CK}	Clock Period	RL=6	12		ns
			RL=5	15		ns
			RL=4	18		ns
			RL=3	30		ns
t_{KHKL} t_{KLKH}	t_{CKH} t_{CKL}	Clock High to Clock Low Clock Low to Clock High		3.5		ns
t_F, t_R(1)	t_{CKT}	Clock Rise Time Clock Fall Time			1.5	ns

1. Clock Rise and Fall times are defined between V_{IH} min. and V_{IL} Max.

Table 15. Synchronous mode - Address Latch AC characteristics

Symbol	Alt	Description		Min	Max	Unit
t_{AVLL}(1)	t_{ASVL}	Address Valid to $\overline{L}$ Low		– 3		ns
t_{AVEL}(1)	t_{ASCL}	Address Valid to Chip Enable Low		– 3		ns
t_{LHAX}	t_{AHV}	$\overline{L}$ High to Address Transition		0		ns
t_{LLLH}(2)	t_{VPL}	$\overline{L}$ Low to $\overline{L}$ High		7		ns
t_{LLKH}(3)	t_{VSCK}	$\overline{L}$ Low to Clock High	RL=5, 6	3.5		ns
			RL=3, 4	5		ns
t_{ELKH}(3)	t_{CLCK}	Chip enable Low to Clock High	RL=5, 6	3.5		ns
			RL=3, 4	5		ns
t_{KHLL}, t_{KHEL}	t_{VHVL}	$\overline{L}$ High Hold Time from Clock Rising Edge. Chip Enable High Hold Time from Clock Rising Edge.		TBD(4)		ns
t_{KHLH}(3)	t_{CKVH}	Larch Enable Low Hold Time from Clock		2.5		ns

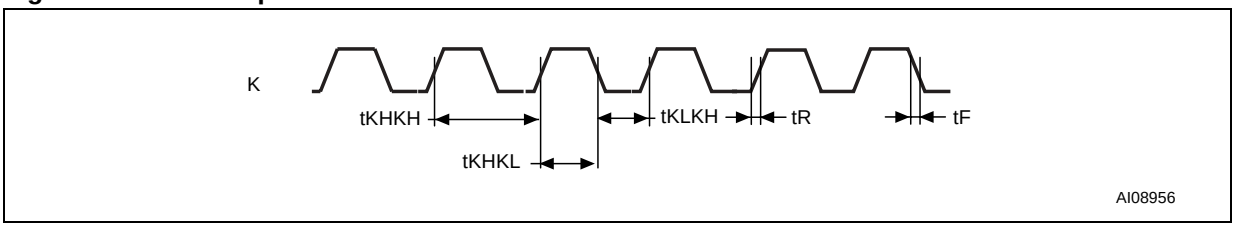
1. t_{AVEL} is applicable if $\overline{E1}$ is brought to Low after $\overline{L}$ is brought to Low with t_{LLEL} time respected. Both t_{AVEL} and t_{AVLL} must be respected if t_{LLEL} is not respected.
2. t_{LLLH} is defined from the falling edge of either $\overline{E1}$ or $\overline{L}$ whichever comes last. At least one valid Clock edge must occur when $\overline{L}$ is Low.
3. Applicable to the first valid clock edge.
4. TBD stands for To Be Defined.

Table 16. Synchronous Burst Read AC characteristics

Symbol	Alt	Description		Min	Max	Unit
t_{ELEH}, t_{LLLL}	t_{RCB}	Burst Read Cycle Time Address Valid Low to Address Valid Low			8000	ns
t_{KHQV}(1)	t_{AC}	Clock Access Time	RL=5, 6		8	ns
			RL=3, 4		10	ns
t_{KHQX}(1)	t_{CKQX}	Output Hold Time from Clock		3		ns
t_{ELTL}(1)	t_{CLTL}	Chip Enable Low to $\overline{WAIT}$ Low		5	20	ns
t_{LLTL}(1)	t_{VLTL}	Latch Enable Low to $\overline{WAIT}$ Low		5	20	ns
t_{ELQX}(2)	t_{CLZ}	Chip Enable Low to Output Transition		5		ns
t_{GLTL}(1)	t_{OLTL}	Output Enable Low to $\overline{WAIT}$ Low		0	20	ns
t_{GLQX}(2)	t_{OLZ}	Output Enable Low to Output Transition		10		ns
t_{KHTL}, t_{KHTH}(1)	t_{CKTV}, t_{CKTX}	Clock high to $\overline{WAIT}$ High		3	12	ns
t_{GLKH}	t_{OLQ}	Output Enable Low to Clock High		30		ns
t_{BLKH}(3)	t_{BLQ}	$\overline{LB}$, $\overline{UB}$ High to Clock High		30		ns
t_{GHQZ}	t_{OHZ}	Output Enable High to Output Hi-Z			20	ns
t_{EHQZ}, t_{EHTZ}	t_{CHZ}	Chip Enable High to Output Hi-Z Chip Enable High to $\overline{WAIT}$ Hi-Z			20	ns
t_{GHTZ}(1)	t_{OHTZ}	Output Enable High to $\overline{WAIT}$ Hi-Z			12	ns
t_{GHKH}	t_{OSCK}	Output Enable High to Clock High		5		ns
t_{KHGH}	t_{CKOH}	Output Enable Low Hold Time from Clock High		5		ns
t_{KHEH}	t_{CKCLH}	Chip Enable Low Hold Time from Clock High		5		ns
t_{KHBH}	t_{CKBH}	$\overline{LB}$, $\overline{UB}$ Low Hold Time from Clock High		5		ns
t_{EHEL}	t_{CP}	Chip Enable High to Chip Enable Low		15		ns
t_{EHLL}(4)	t_{TRB}	Burst Terminate Recovery Time	Burst Length = 8, 16 Words	30		ns
			Burst Length = Continuous	70		ns

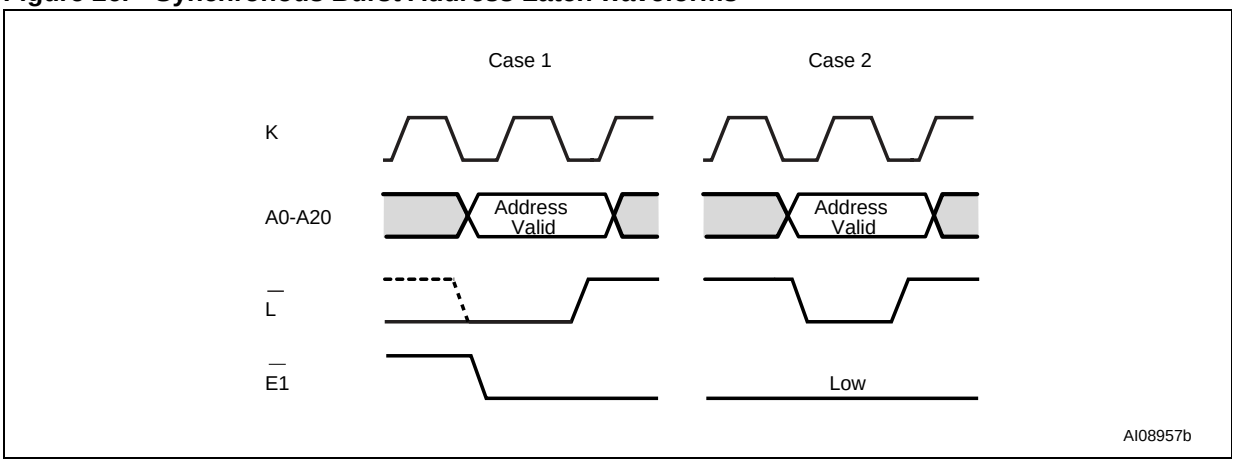
1. See *Figure 25*.
2. The output Load capacitance is 5pF without any other load.
3. Once set, it must not change before the end of the Burst operation.
4. Defined from the rising edge of $\overline{E1}$ to the falling edge of either $\overline{L}$ or $\overline{E1}$ whichever occurs last.

Figure 25. Clock Input Waveform



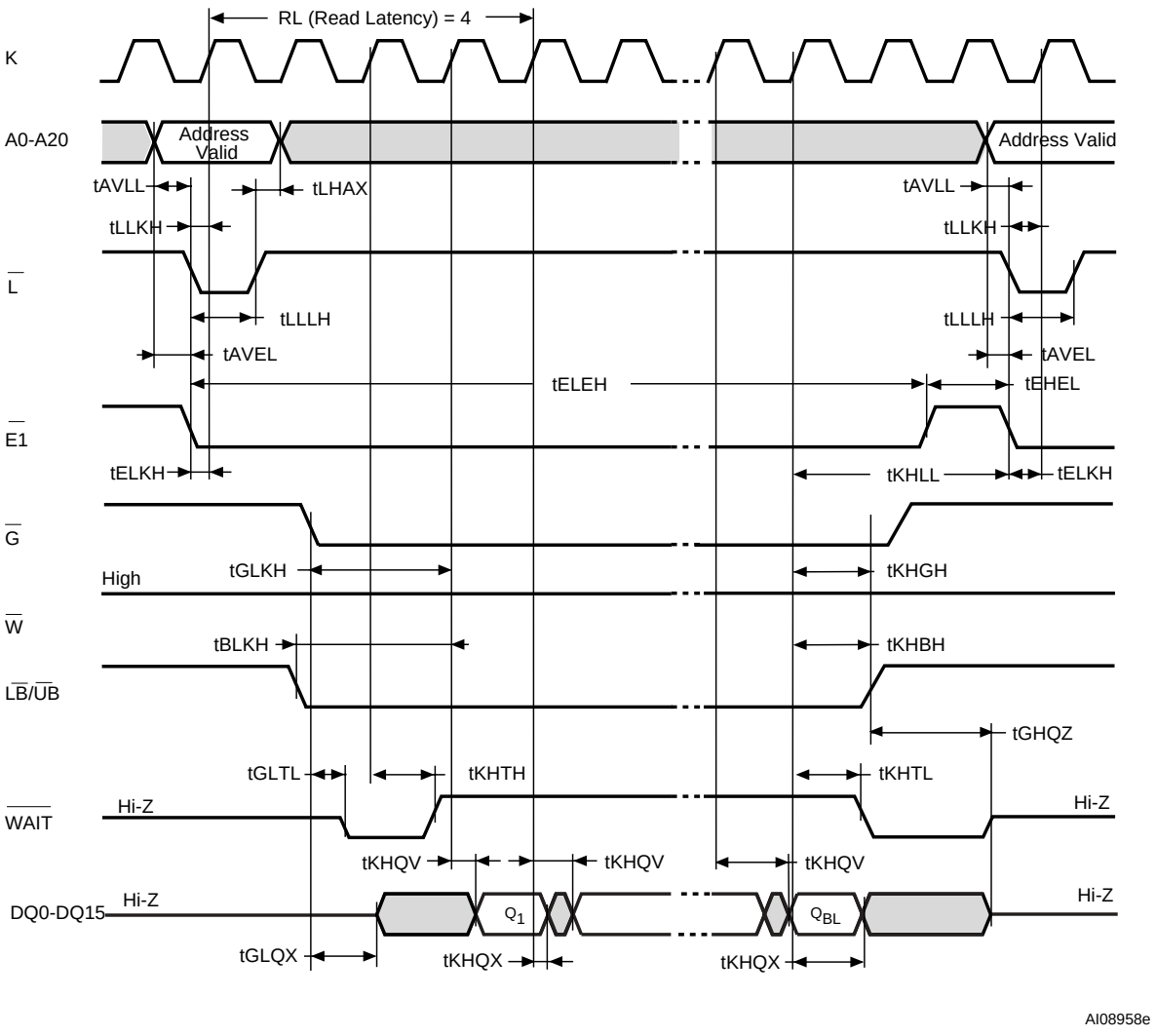
1. The Clock K input must be available and stable before the first Read or Write access to the memory following a Set CR Register Sequence. It is prohibited to change the clock frequency during a Read or Write access to the memory.
2. $t_{F,}$ t_R is defined between V_{IH} min. and V_{IL} max.

Figure 26. Synchronous Burst Address Latch waveforms



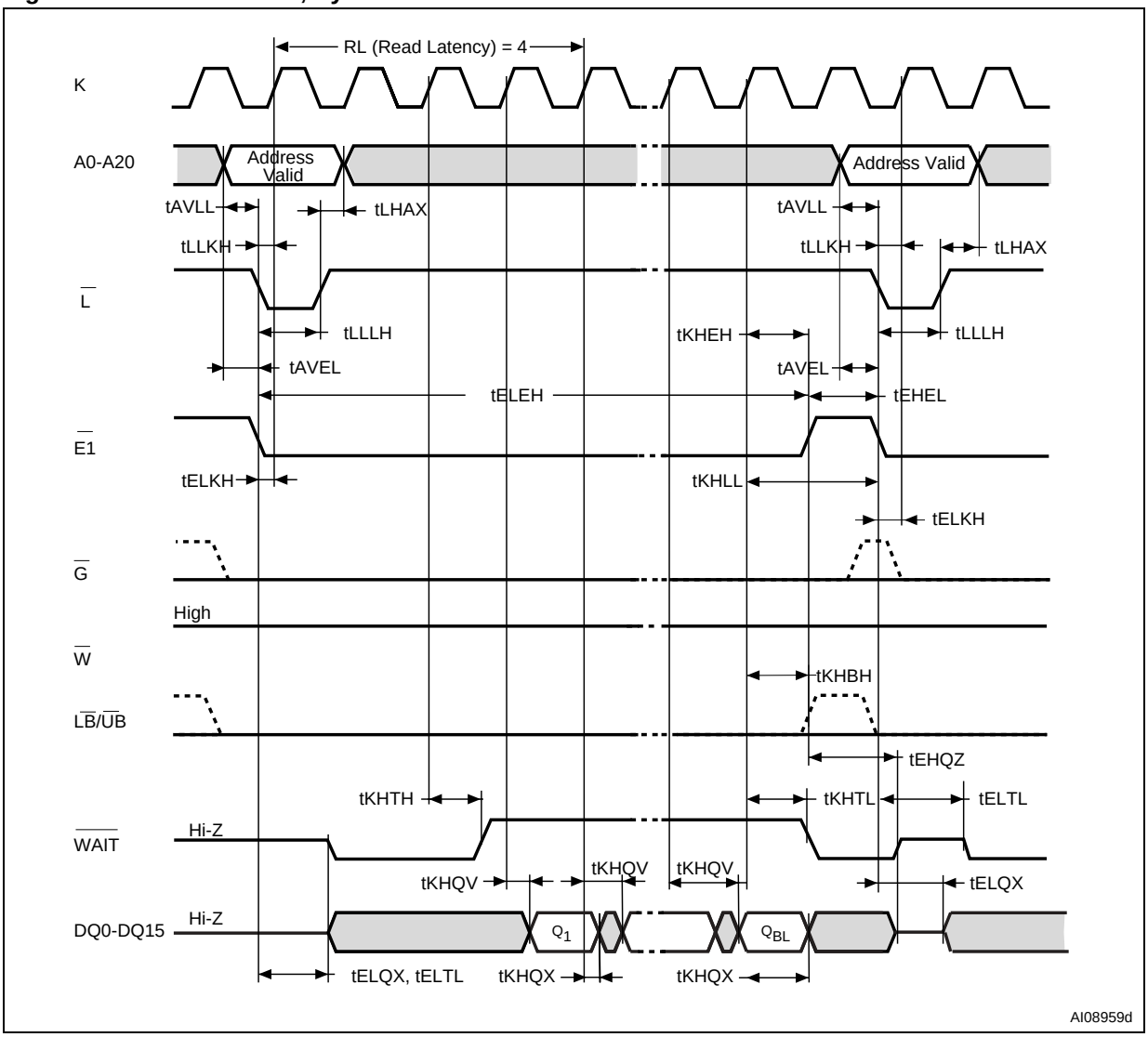
1. Case 1 describes the Address Latch waveform when $\overline{E1}$ is driven Low after $\overline{L}$ is driven Low.
   Case 2 describes the Address Latch waveform when $\overline{L}$ is driven Low after $\overline{E1}$ is driven Low.
2. t_{LLLH} is defined from the falling edge of $\overline{E1}$ or $\overline{L}$ whichever occurs first.
   At least one valid clock edge must occur during $\overline{L}$ low.
3. t_{LLKH} and t_{ELKL} apply to the first valid clock edge during $\overline{L}$ Low.

Figure 27. $\overline{G}$ controlled, Synchronous Read AC waveforms



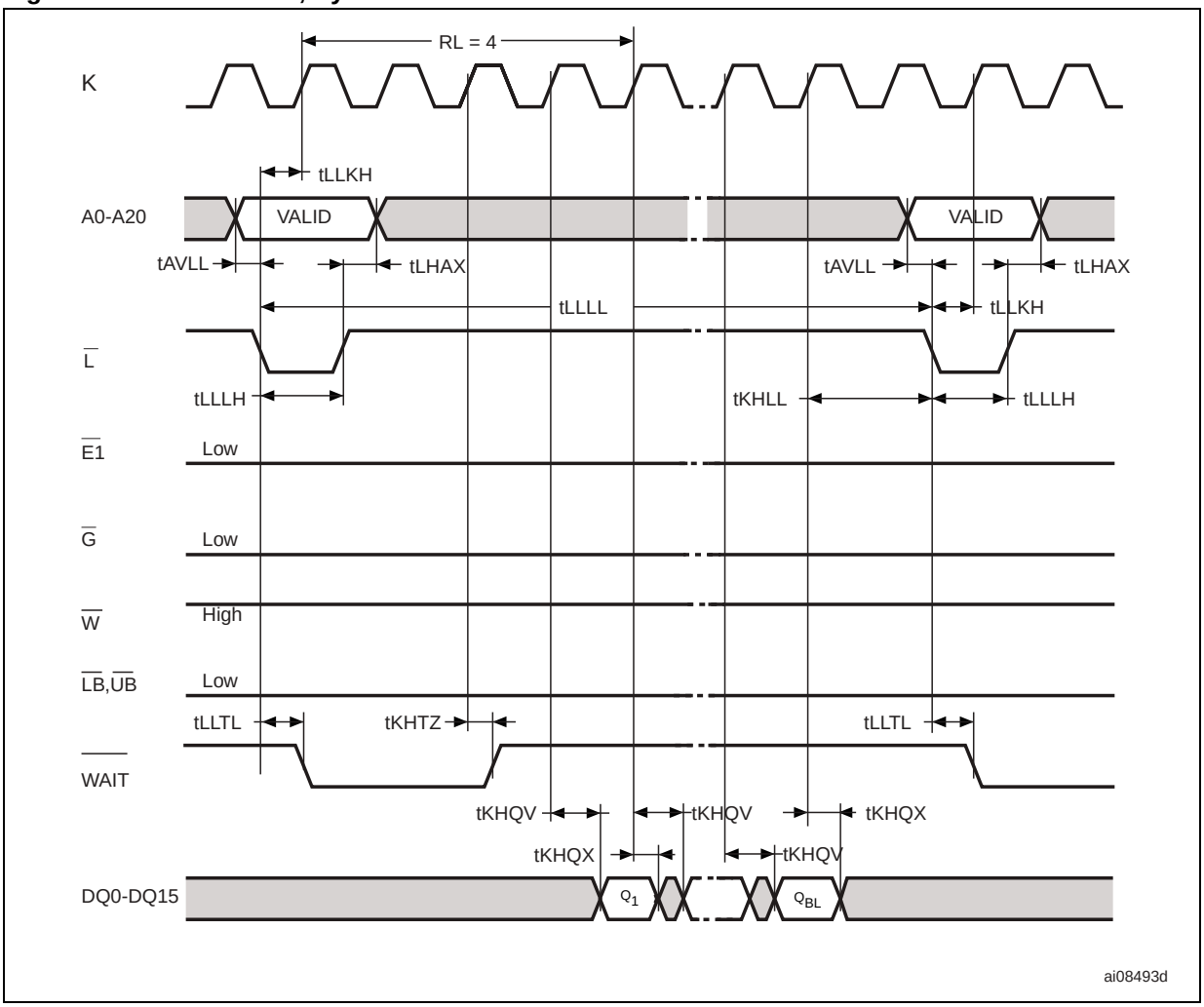
1. The above diagram is an example of Synchronous Burst Read waveforms for a Latency of 4 clock cycles, and a Burst Length of 8 or 16 Words. Chip Enable E2 is held High, V_{IH}.

Figure 28. $\overline{E1}$ controlled, Synchronous Read AC waveforms



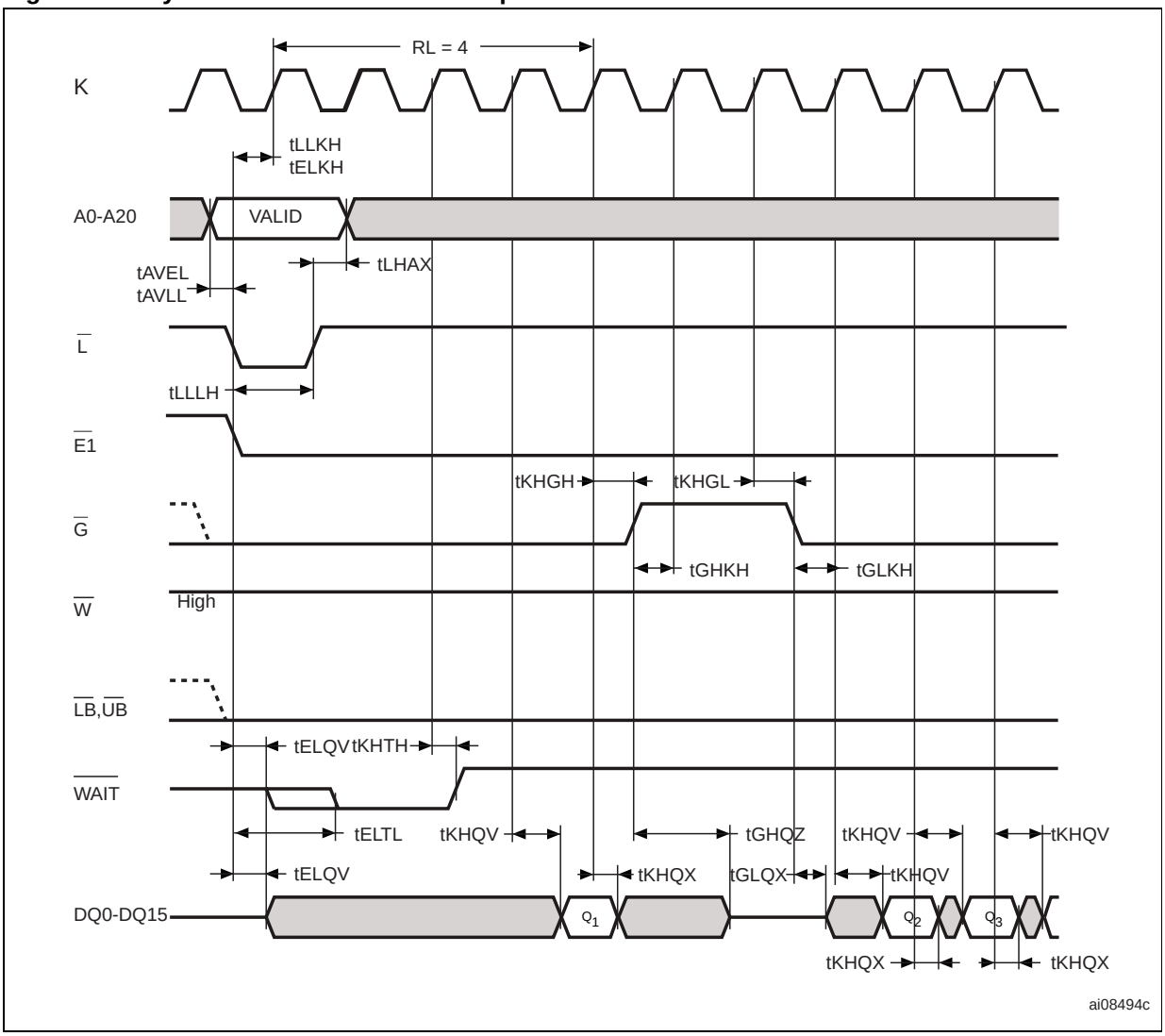
1. The above diagram is an example of Synchronous Burst Read waveforms for a Latency of 4 clock cycles and a Burst Length of 8 or 16 Words. Chip Enable E2 is held High, V_{IH}

Figure 29. $\overline{L}$ controlled, Synchronous Read AC waveforms



1. The above diagram is an example of Synchronous Burst Read waveforms for a Latency of 4 clock cycles.

Figure 30. Synchronous Burst Read Suspend and Resume AC Waveforms



1. The above diagram is an example of Synchronous Burst Read Suspend and Resume waveforms for a Latency of 4 clock cycles.

Figure 31. Synchronous Burst Read Terminate AC waveforms

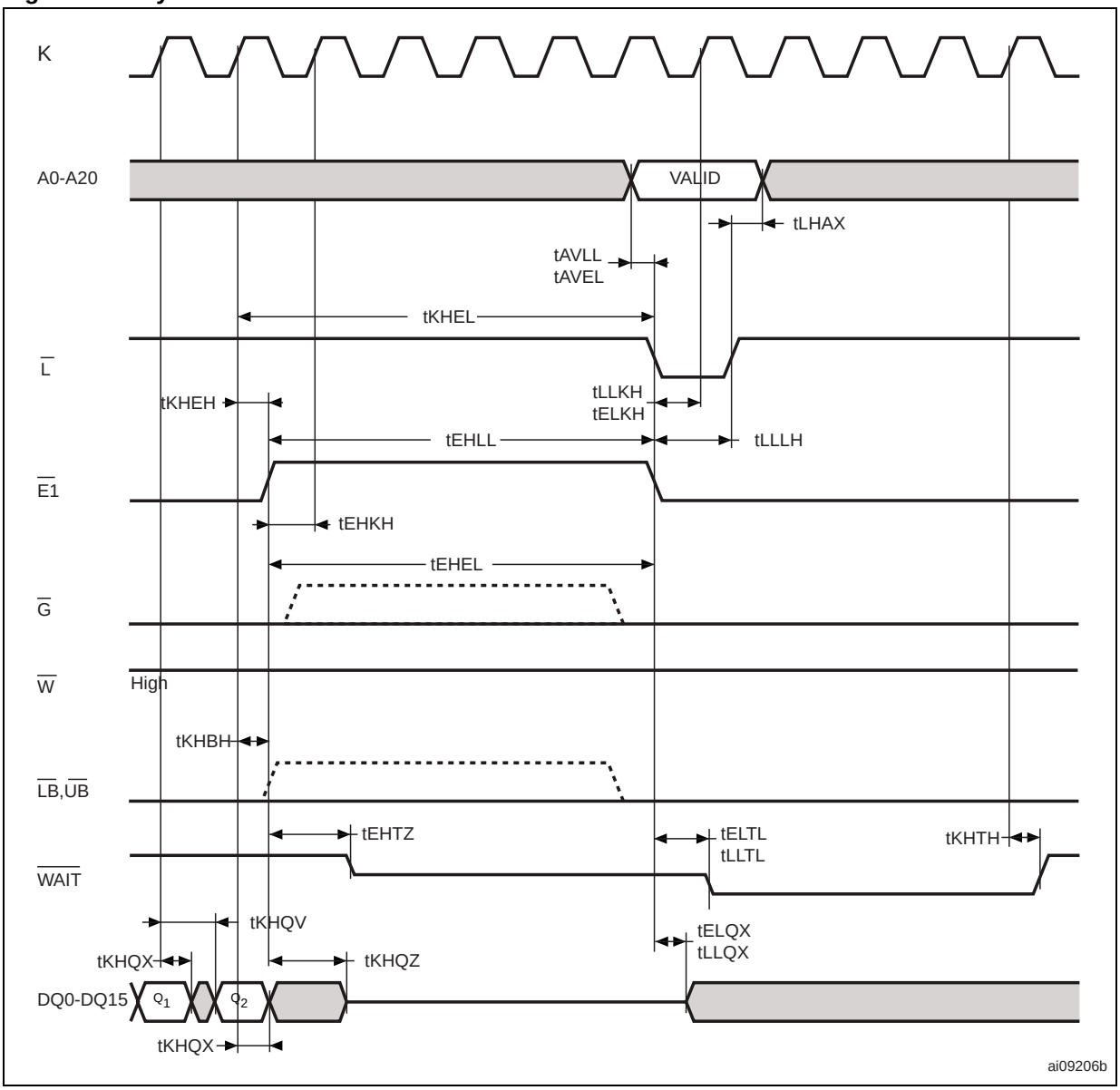
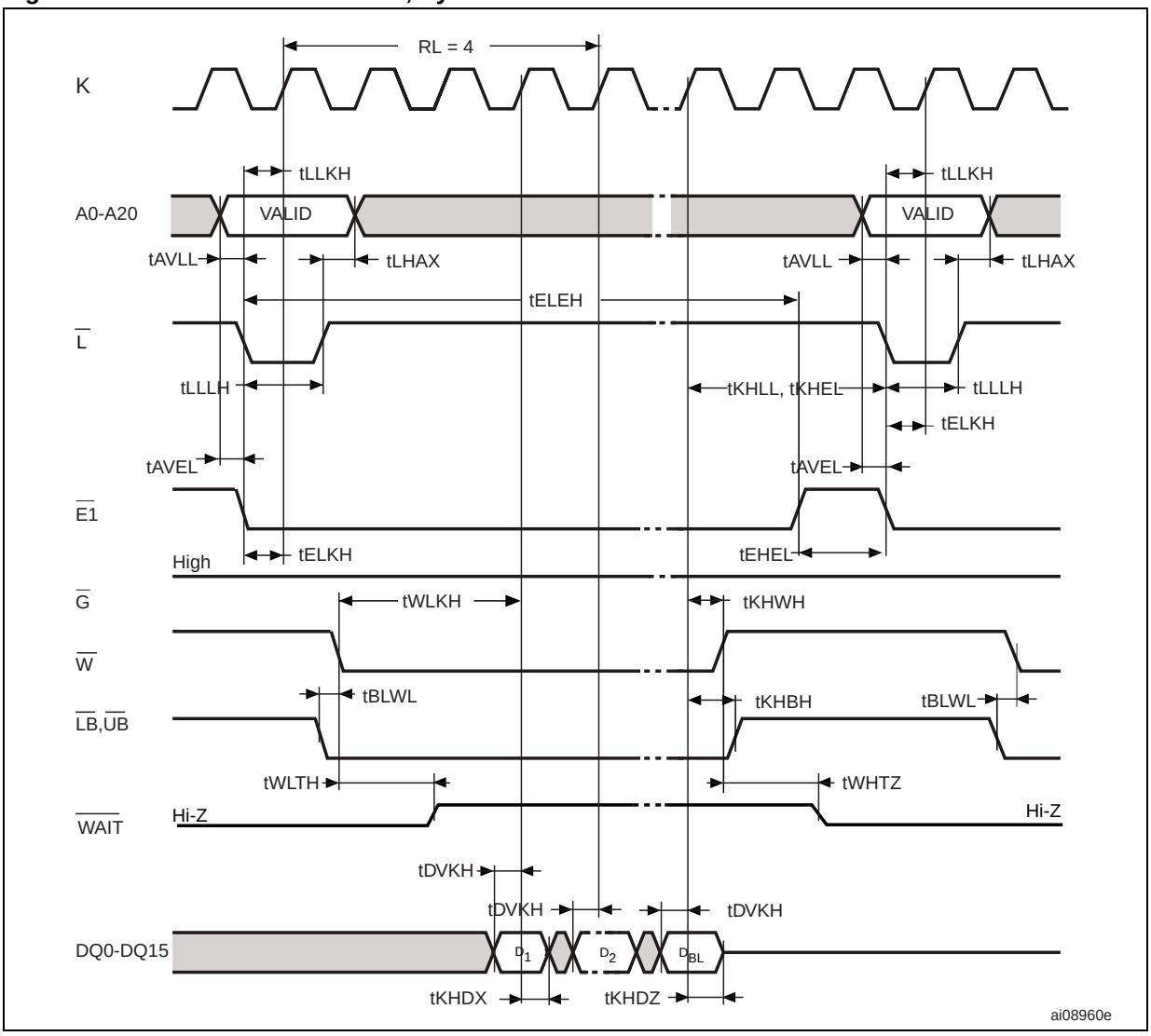


Table 17. Synchronous Burst Write AC characteristics

Symbol	Alt	Description		Min	Max	Unit
t_{ELEH} t_{LLLL} t_{LLEH}	t_{WCB}	Burst Write Cycle Time			8	µs
t_{DVKH}	t_{DSCK}	Input Valid to Clock High Clock High to Input Valid		7		ns
t_{KHDZ}, t_{KHDX}	t_{DHCK}	Clock high to Input Transition or Input High Z		3		ns
t_{WLDV}	t_{WLD}	Write Enable Low to First Input Valid		30		ns
t_{BLEL}, t_{BLLL}, t_{BLWL} (1)	t_{BS}	$\overline{LB}$, $\overline{UB}$ Low to Chip Enable Low $\overline{LB}$, $\overline{UB}$ Low to Address Valid Low $\overline{LB}$, $\overline{UB}$ Low to Write Enable Low		– 5		ns
t_{WLKH1}, t_{WHKH}	t_{WSCK}	Write Enable Setup Time to Clock High		5		ns
t_{KHWL}, t_{KHWH}	t_{CKWH}	Write Enable Hold Time (Low, High) from Clock High		5		ns
t_{KHEH}	t_{CKCLH}	Chip Enable Low Hold Time from Clock High		5		ns
t_{EHKH}	t_{CHCK}	Chip Enable High to Clock High		5		ns
t_{KHBH}	t_{CKBH}	$\overline{LB}$, $\overline{UB}$ Low Hold Time from Clock High		5		ns
t_{KHEL} (2)	t_{WRB}	Burst Write Recovery Time		30		ns
t_{EHLL} (3)	t_{TRB}	Burst Terminate Recovery Time	Burst Length = 8, 16 Words	30		ns
			Burst Length = Continuous	70		ns
t_{EHEL}	t_{CP}	Chip Enable High to Chip Enable Low		15		ns
t_{WLTH}	t_{WLTH}	Write Enable Low to $\overline{WAIT}$ High		0	20	ns
t_{WHTZ}	t_{WHTZ}	Write Enable High to $\overline{WAIT}$ High-Z			20	ns
t_{EHTZ}	t_{CHTZ}	Chip Enable High to $\overline{WAIT}$ High-Z			12	ns
t_{ELTH}	t_{CLTH}	Chip Enable E1 Low to $\overline{WAIT}$ High		5	15	ns

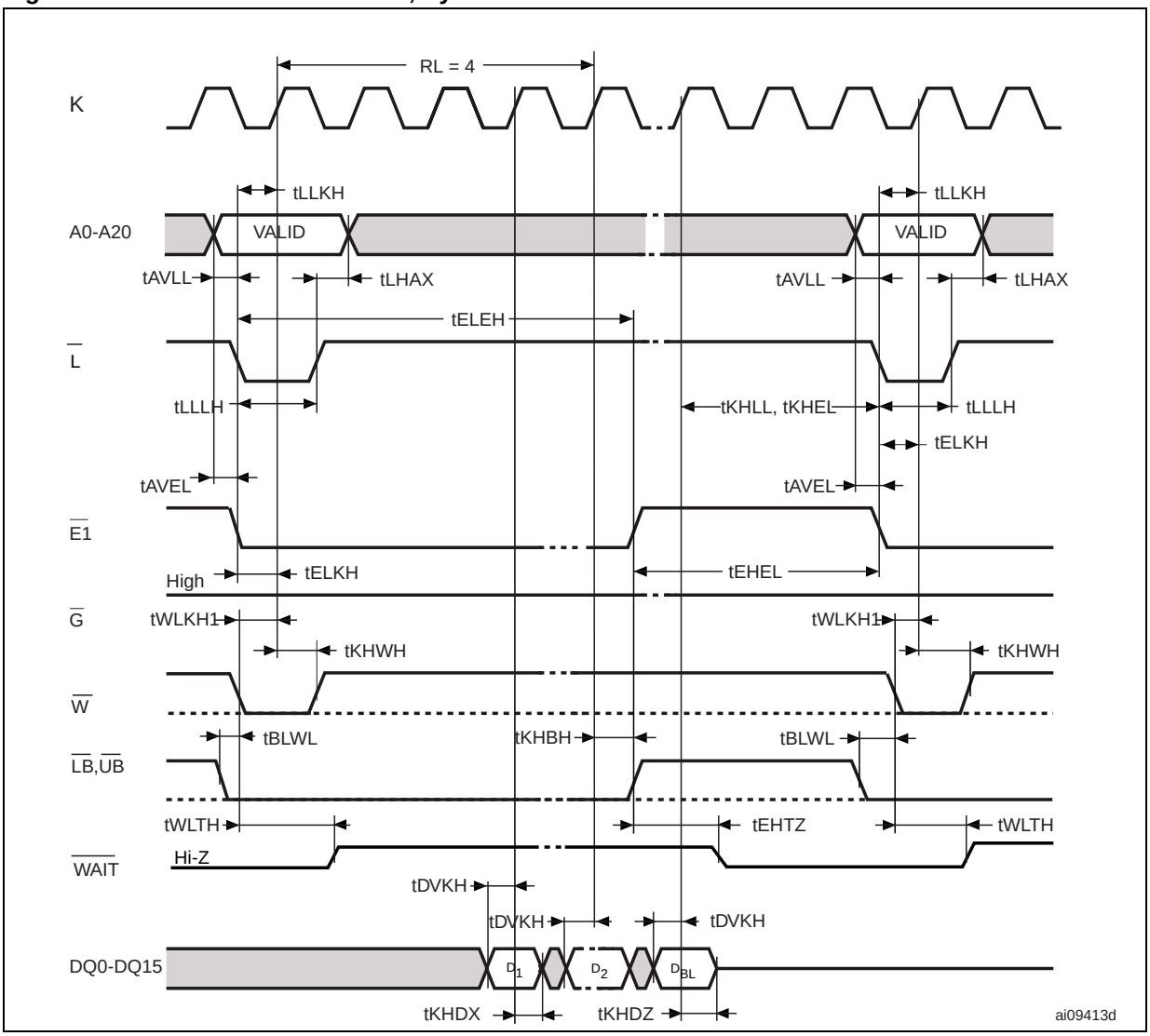
1. Defined from the valid input edge to the falling edge of either $\overline{L}$, $\overline{E1}$ or $\overline{W}$ whichever occurs last. Once set, it must not change before the end of the Burst operation.
2. Defined from the rising edge of the Clock K where the last data inputs have been latched at the end of the Burst Write operation to the falling edge of either $\overline{L}$ or $\overline{E1}$ whichever occurs last for the next access.
3. Defined from the rising edge of $\overline{E1}$ to the falling edge of either $\overline{L}$ or $\overline{E1}$ whichever occurs last for the next access.

Figure 32. $\overline{W}$ Level Control mode, Synchronous Burst Write AC waveforms



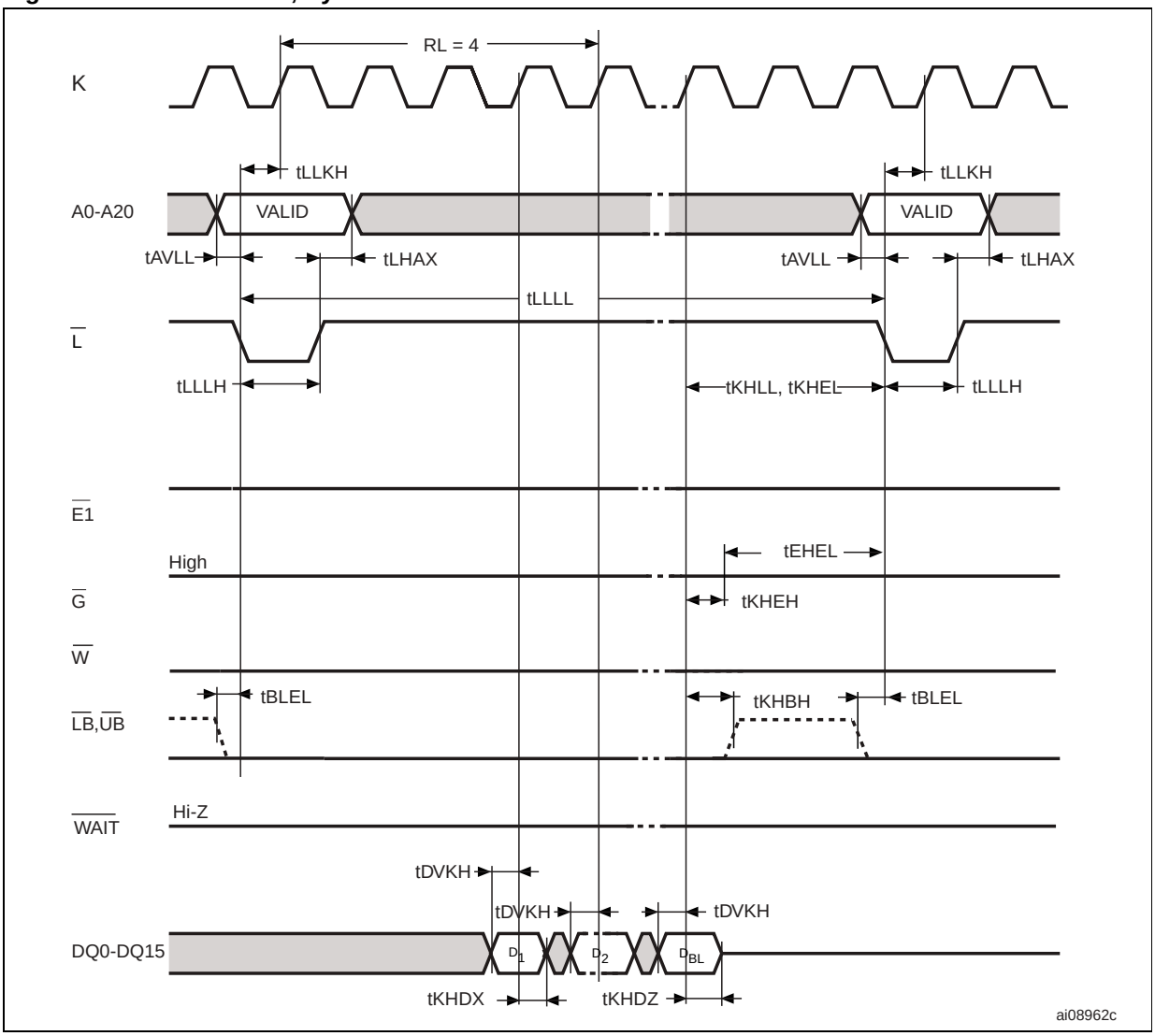
1. The above diagram is an example of Synchronous Burst Write waveforms for a Latency of 4 clock cycles.

Figure 33. $\overline{W}$ Pulse Control mode, Synchronous Burst Write AC waveforms



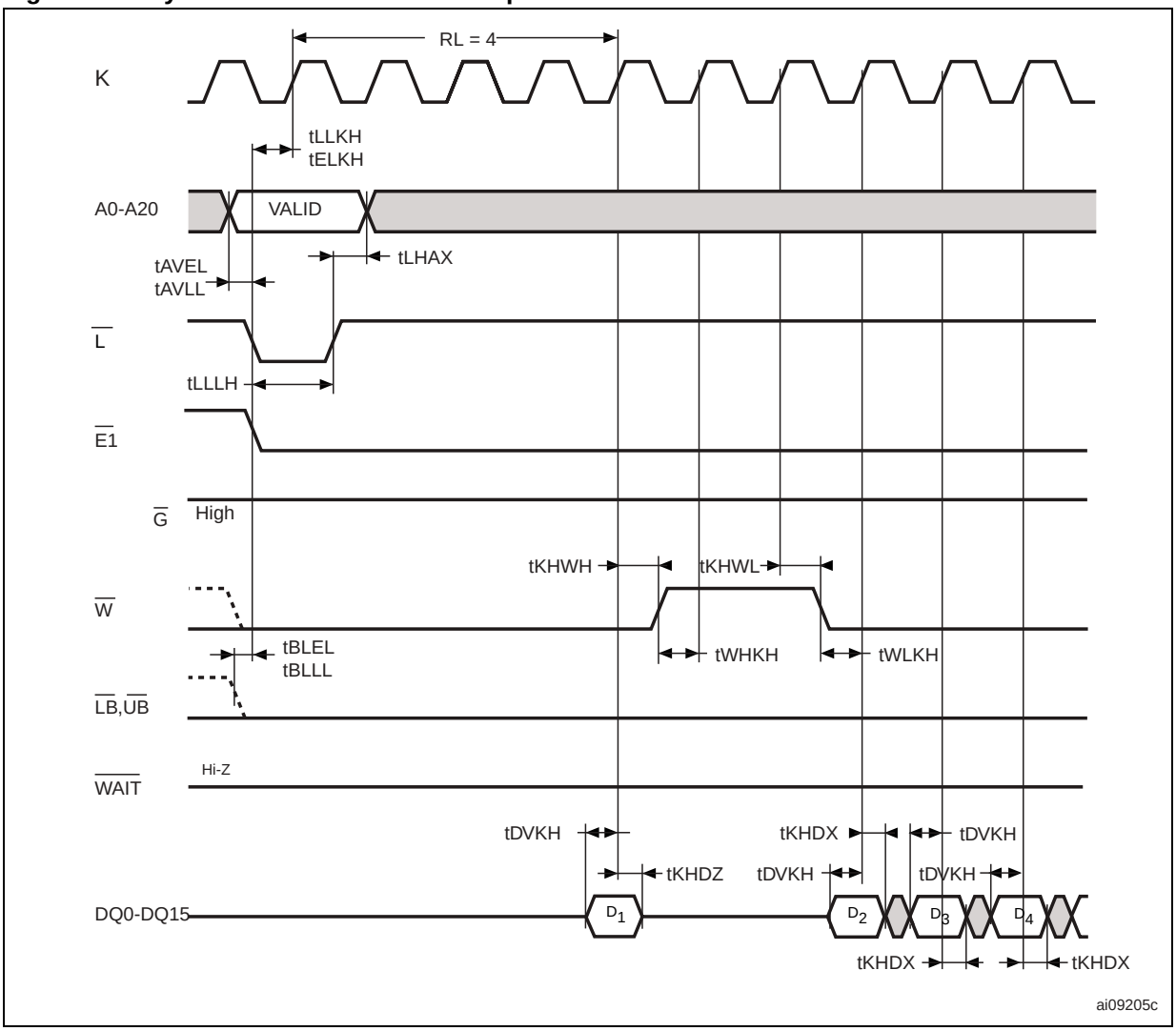
1. The above diagram is an example of Synchronous Burst Write waveforms for a Latency of 4 clock cycles.

Figure 34. $\overline{L}$ Controlled, Synchronous Write AC waveforms



1. The above diagram is an example of Synchronous Burst Write waveforms for a Latency of 4 clock cycles.

Figure 35. Synchronous Burst Write Suspend and Resume AC waveforms



1. The above diagram is an example of Synchronous Burst Write Suspend and Resume waveforms for a Latency of 4 clock cycles.

Figure 36. Synchronous Burst Write Terminate AC waveforms

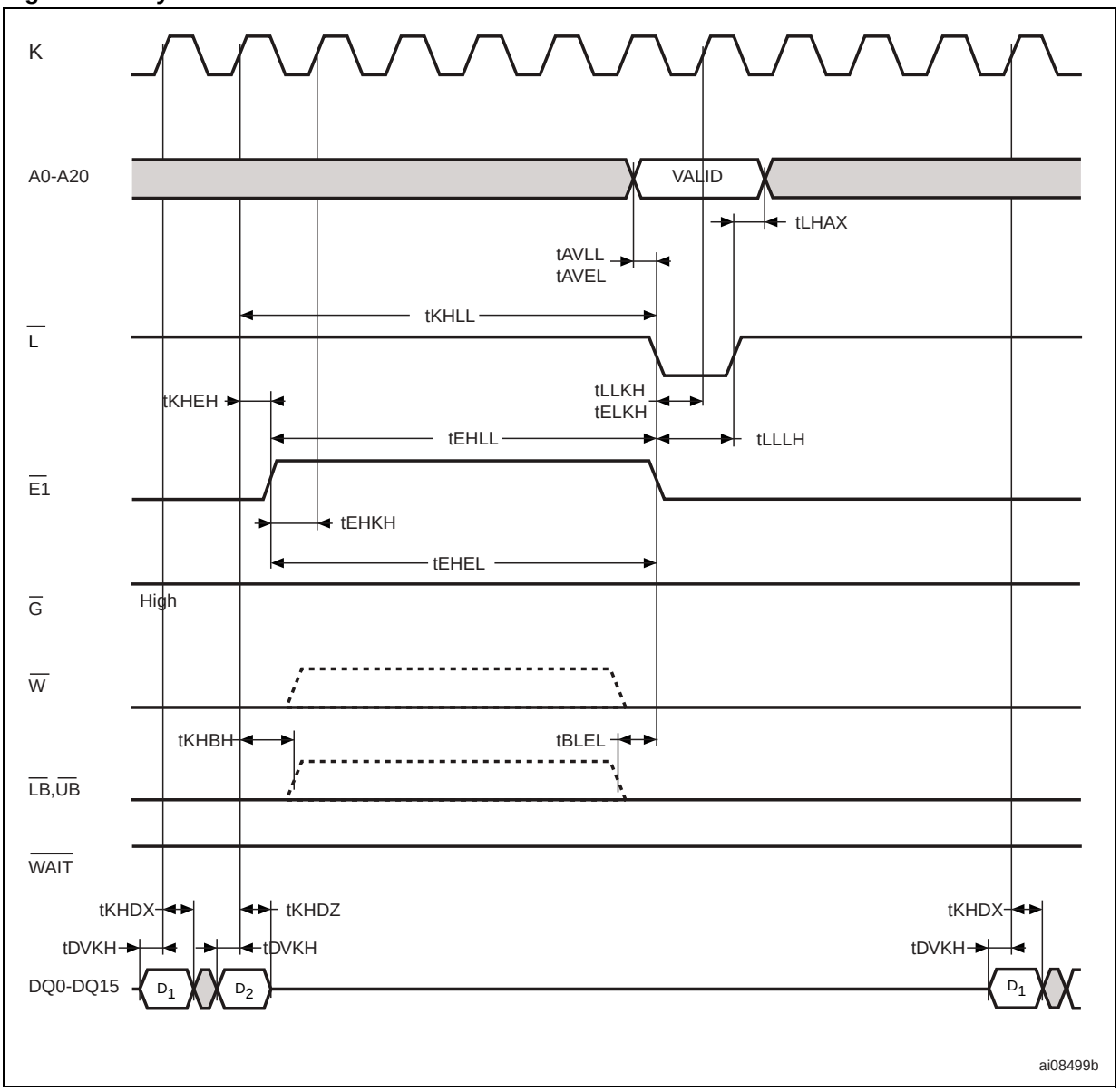
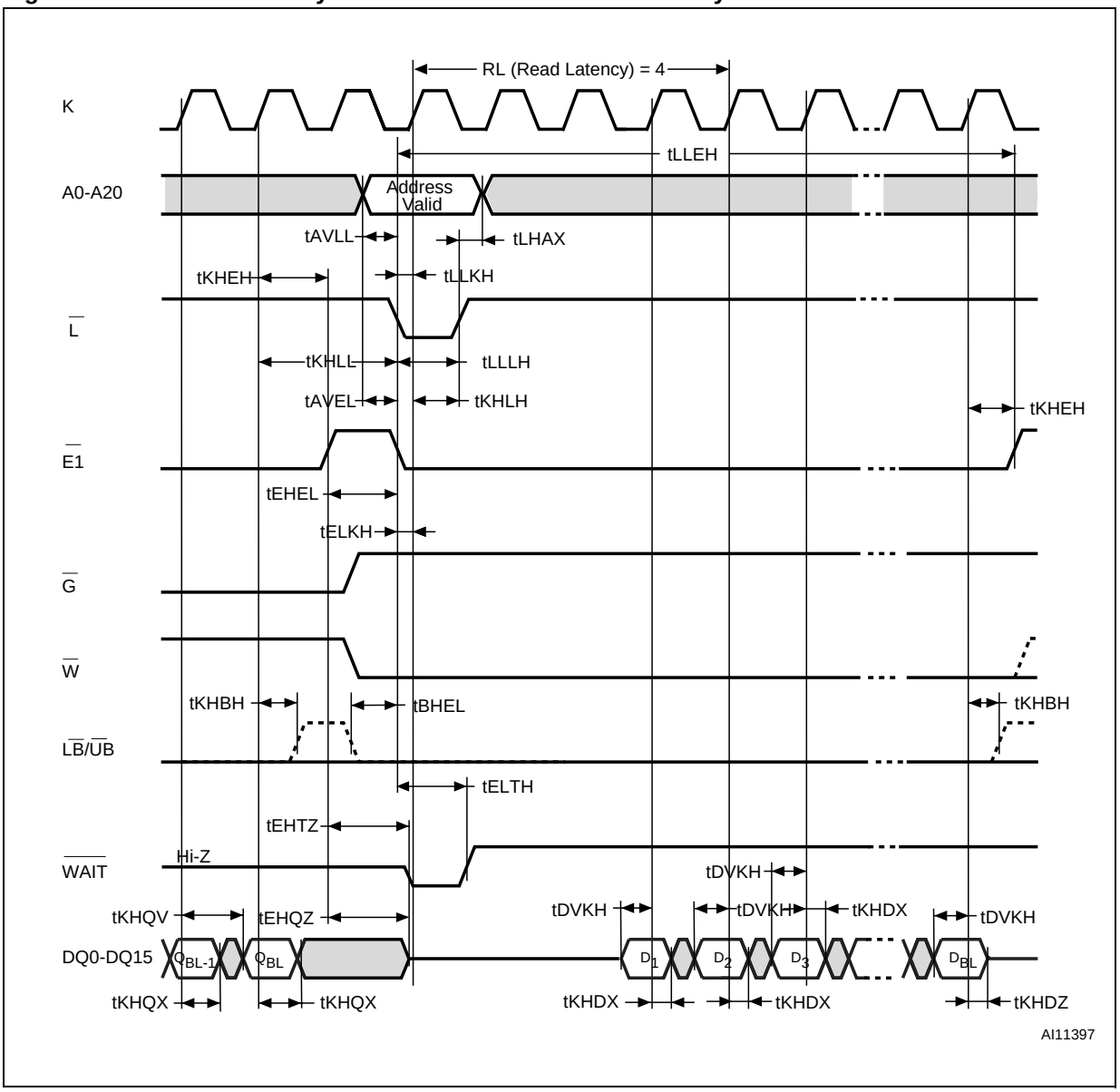
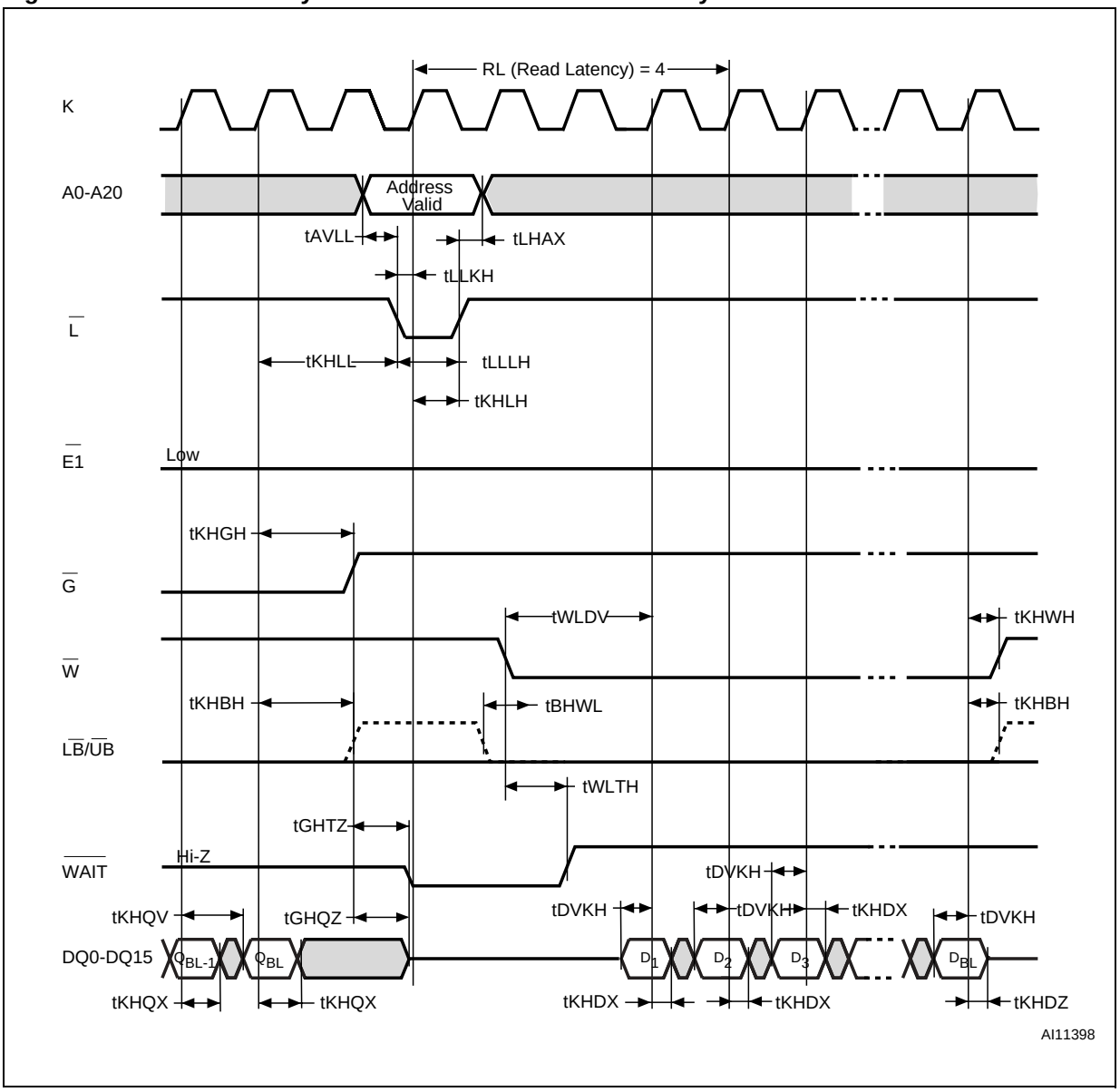


Figure 37. $\overline{E1}$ controlled Synchronous Burst Read followed by Write AC waveforms



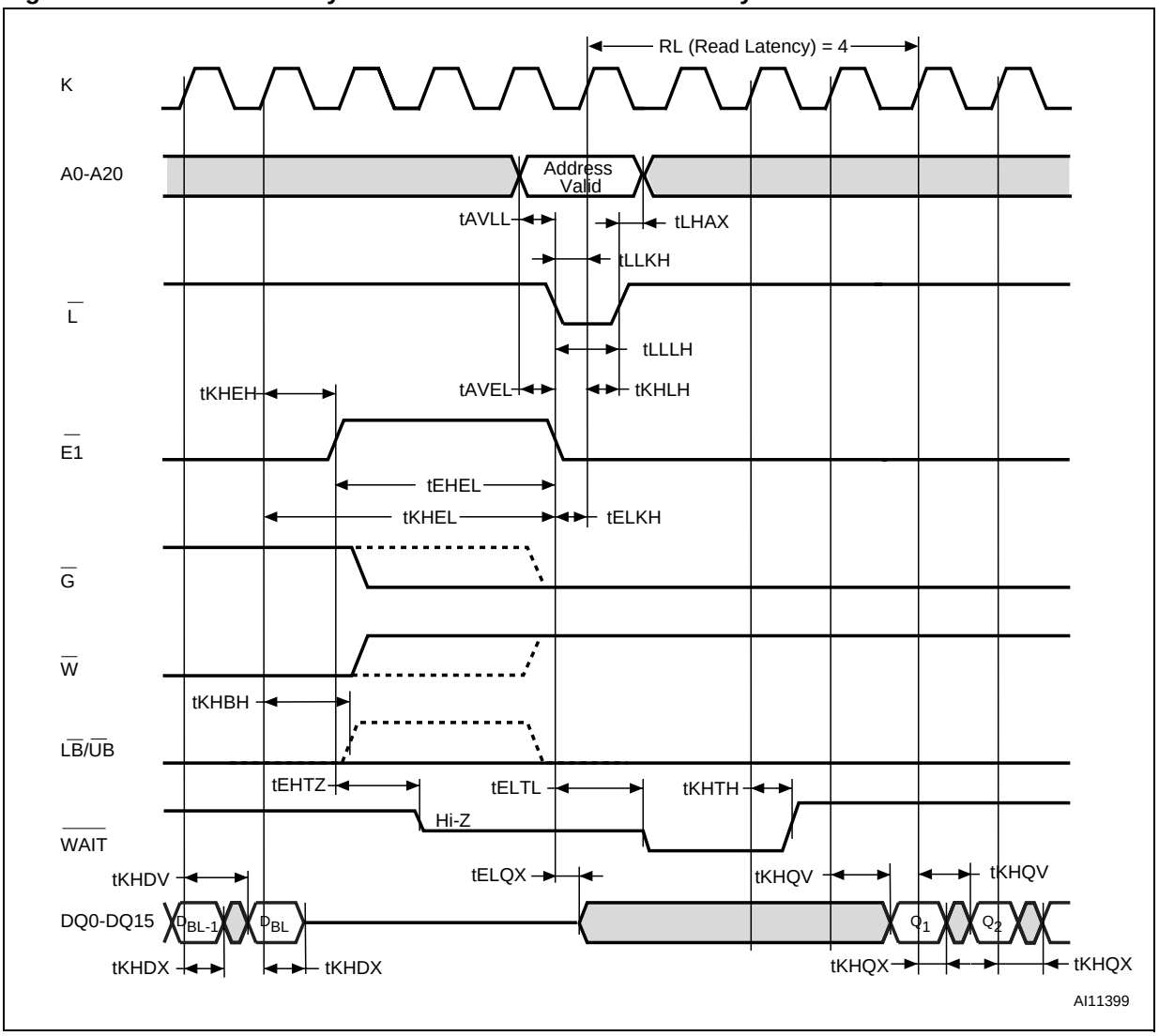
1. The above diagram assumes that E2 is high, V_{IH}, the valid clock edge is the rising edge and the Burst Length is of 8 or 16 Words.

Figure 38. $\overline{L}$ controlled Synchronous Burst Read followed by Write AC waveforms



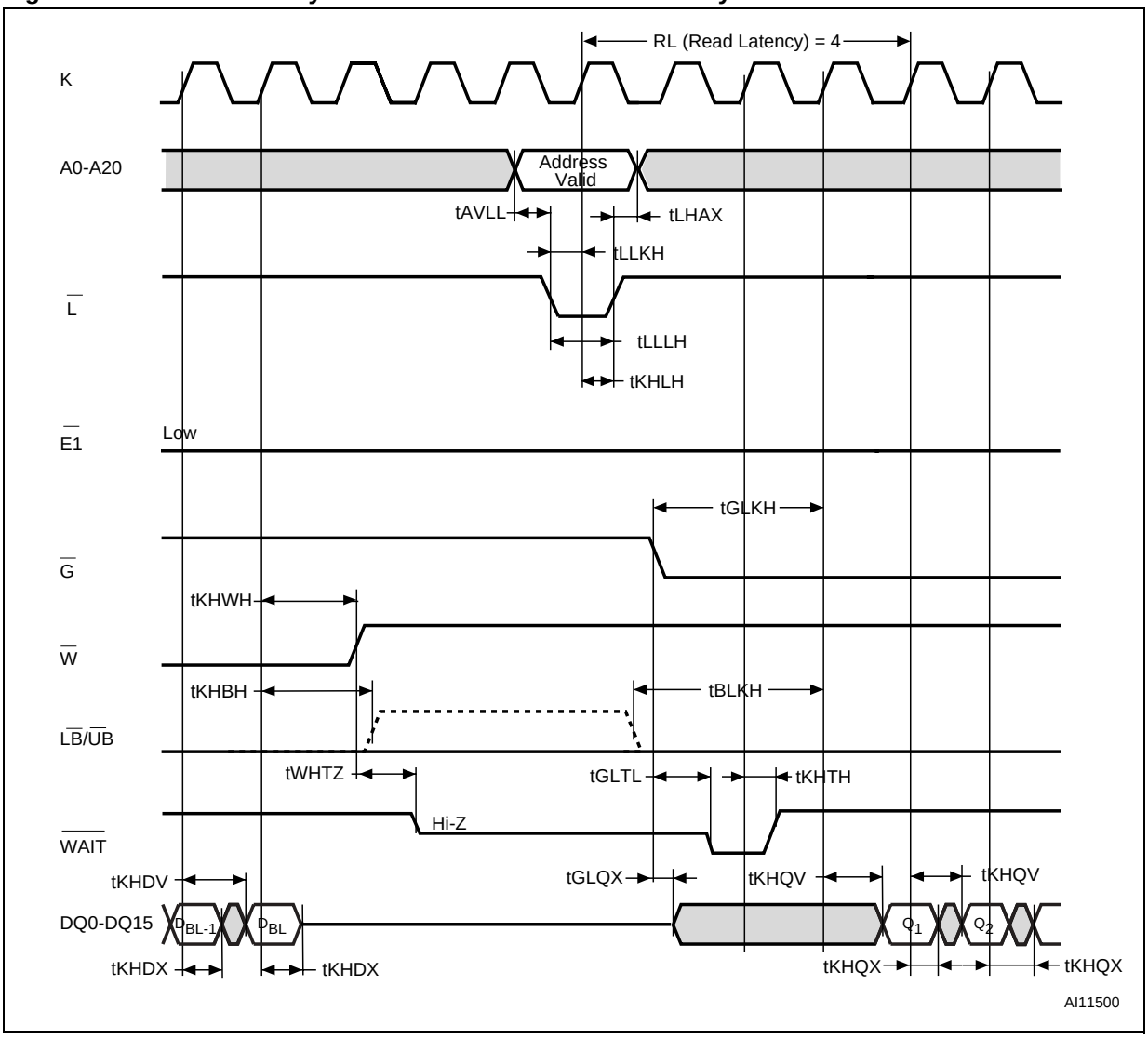
1. The above diagram assumes that E2 is high, V_{IH}, the valid clock edge is the rising edge and the Burst Length is of 8 or 16 Words.

Figure 39. $\overline{E1}$ controlled Synchronous Burst Write followed by Read AC waveforms



1. The above diagram assumes that E2 is high, V_{IH}, the valid clock edge is the rising edge and the Burst Length is of 8 or 16 Words.

Figure 40. $\overline{L}$ controlled Synchronous Burst Write followed by Read AC Waveforms



1. The above diagram assumes that E2 is high, V_{IH}, the valid clock edge is the rising edge and the Burst Length is of 8 or 16 Words.

Table 18. Power-Down and Power-Up AC characteristics

Symbol	Alt.	Parameter	M69KB048BD, M69AB048BD		Unit
			Min	Max	
t_{CLEL}	t_{CSP}	E2 Low Setup Time for Power Down Entry	20		ns
t_{ELCH}(1)	t_{C2LP}	E2 Low Hold Time after Power Down Entry	70		ns
t_{ELCH2}	t_{C2LPR}	E2 Low Hold Time for Reset to Asynchronous Mode	70		ns
t_{EHEL}, t_{CHCL} t_{CHEL}(2)	t_{CHH}	$\overline{E1}$ High Hold Time following E2 High after Power-Up or Power-Down Exit (Deep Power-Down Mode only) or E2 High to $\overline{E1}$ Low	300		µs
t_{CHCL2}(3)	t_{CHHP}	$\overline{E1}$ High Hold Time following E2 High after Power-Down Exit (not in Deep Power-Down Mode)	70		ns
t_{EHCH}(2)	t_{CHS}	$\overline{E1}$ High Setup Time following E2 High after Power-Down Exit	0		ns
t_{VHCH}	t_{C2LH}	E2 Low Hold Time after Power-up	50		µs

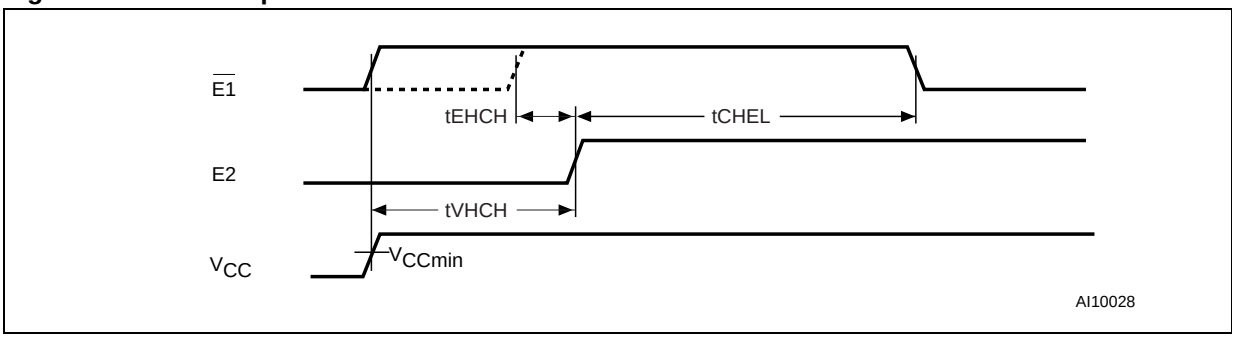
1. Applicable when RP=0 (Reset to Page mode).
2. Applicable also to Power-Up.
3. Applicable when 4 Mbit and 8 Mbit Partial mode is programmed.

Table 19. Standby AC characteristics

Symbol	Alt.	Parameter	M69KB048BD, M69AB048BD		Unit
			Min	Max	
t_{EHGL}	t_{CHOX}	$\overline{E1}$ High to $\overline{G}$ Low for Standby Mode	10		ns
t_{EHWL}(1)	t_{CHWX}	$\overline{E1}$ High to $\overline{W}$ Low for Standby Mode	10		ns
t_{τ}(2)(3)	t_{τ}	Input Transition Time	1	25	ns

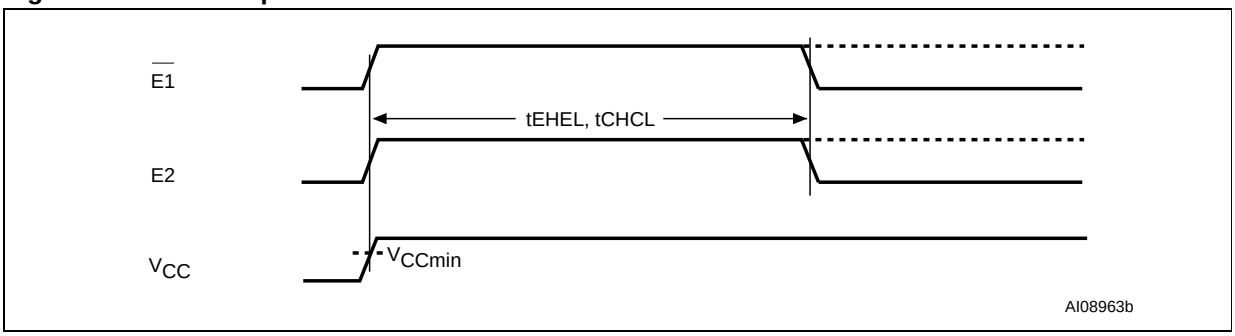
1. Data might be written into any address location if t_{EHWL} minimum value is not respected.
2. Except for Clock input transition time.
3. The input transition , in AC conditions, is 1ns/V (see *Table 9: Operating and AC measurement conditions*). If it is lower than 1ns/V, it may violate AC specification of some timing parameters.

Figure 41. Power-Up AC Waveforms - 1



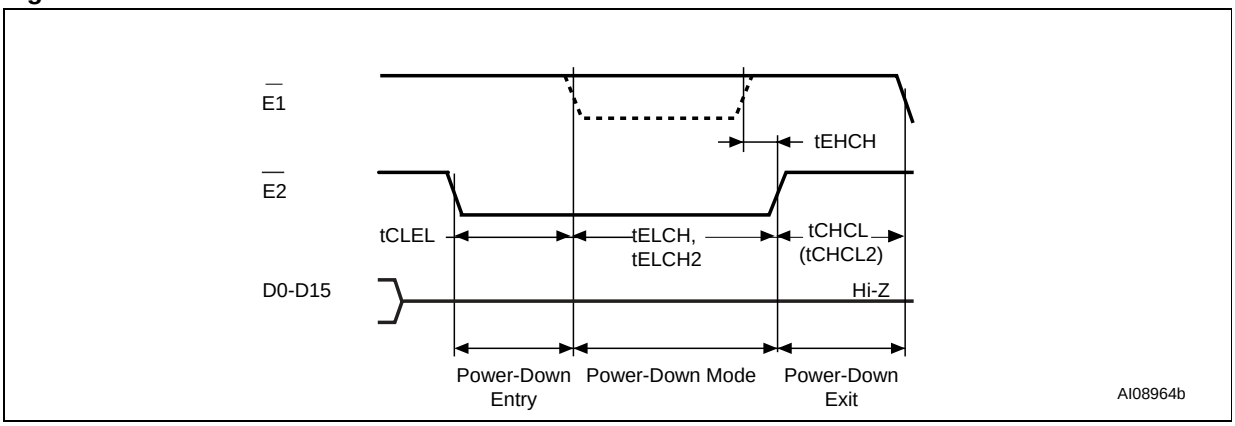
1. t_{E2HE1L} is defined from V_{CC} reaching V_{CC} min to the falling edge of $\overline{E1}$.

Figure 42. Power-Up AC Waveforms - 2



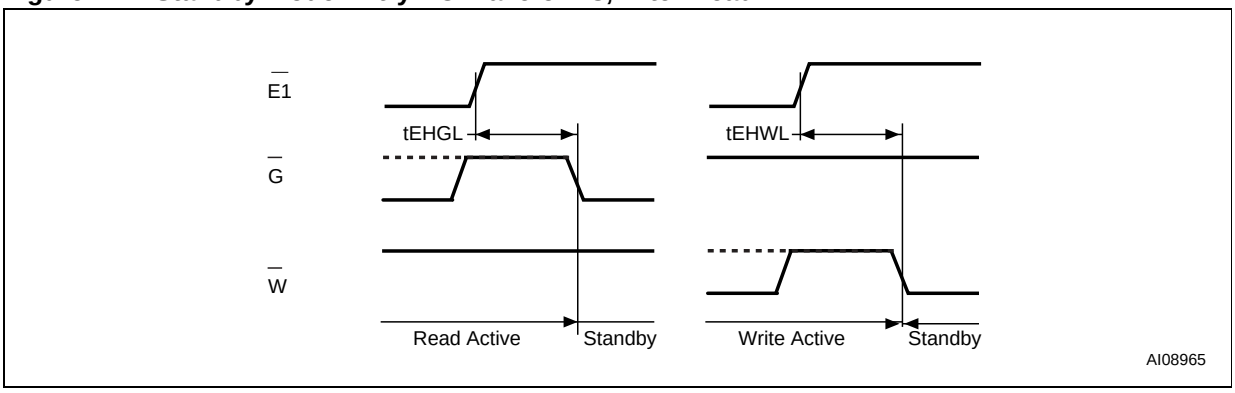
1. t_{EHEV} is defined from V_{CC} reaching V_{CC} min. to $\overline{E1}$, E2 falling edge.

Figure 43. Power Down Mode AC Waveforms



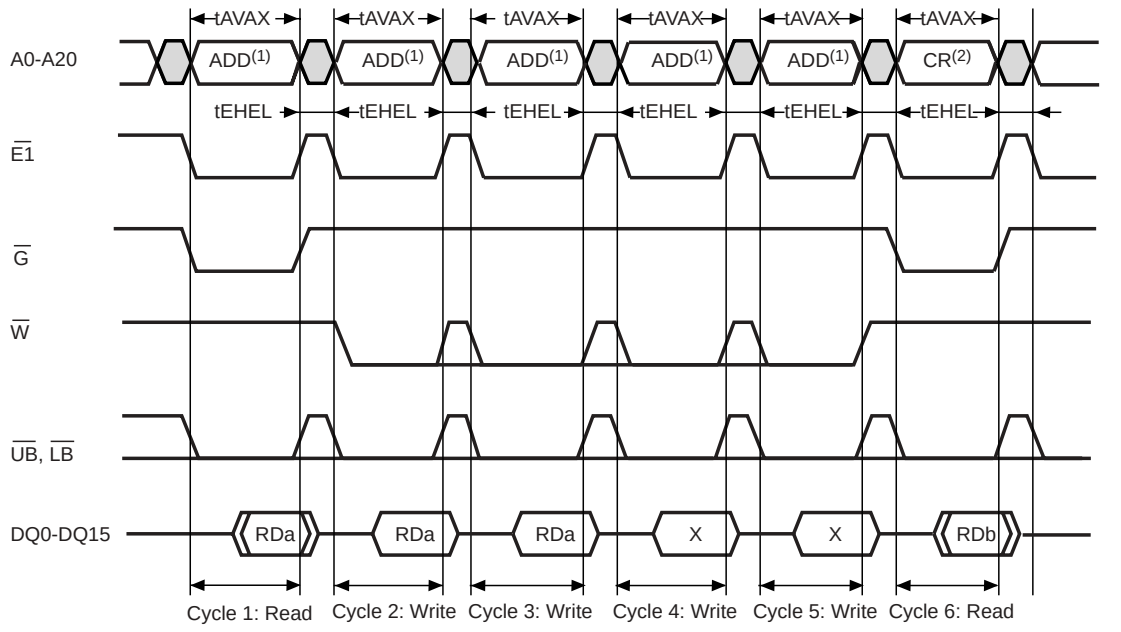
1. Power-Down can also be used as a reset if the Power-Up timing described above (see *Figure 42*) was not respected and if a Power-Down operation has not been performed before this reset.

Figure 44. Standby Mode Entry AC Waveforms, After Read



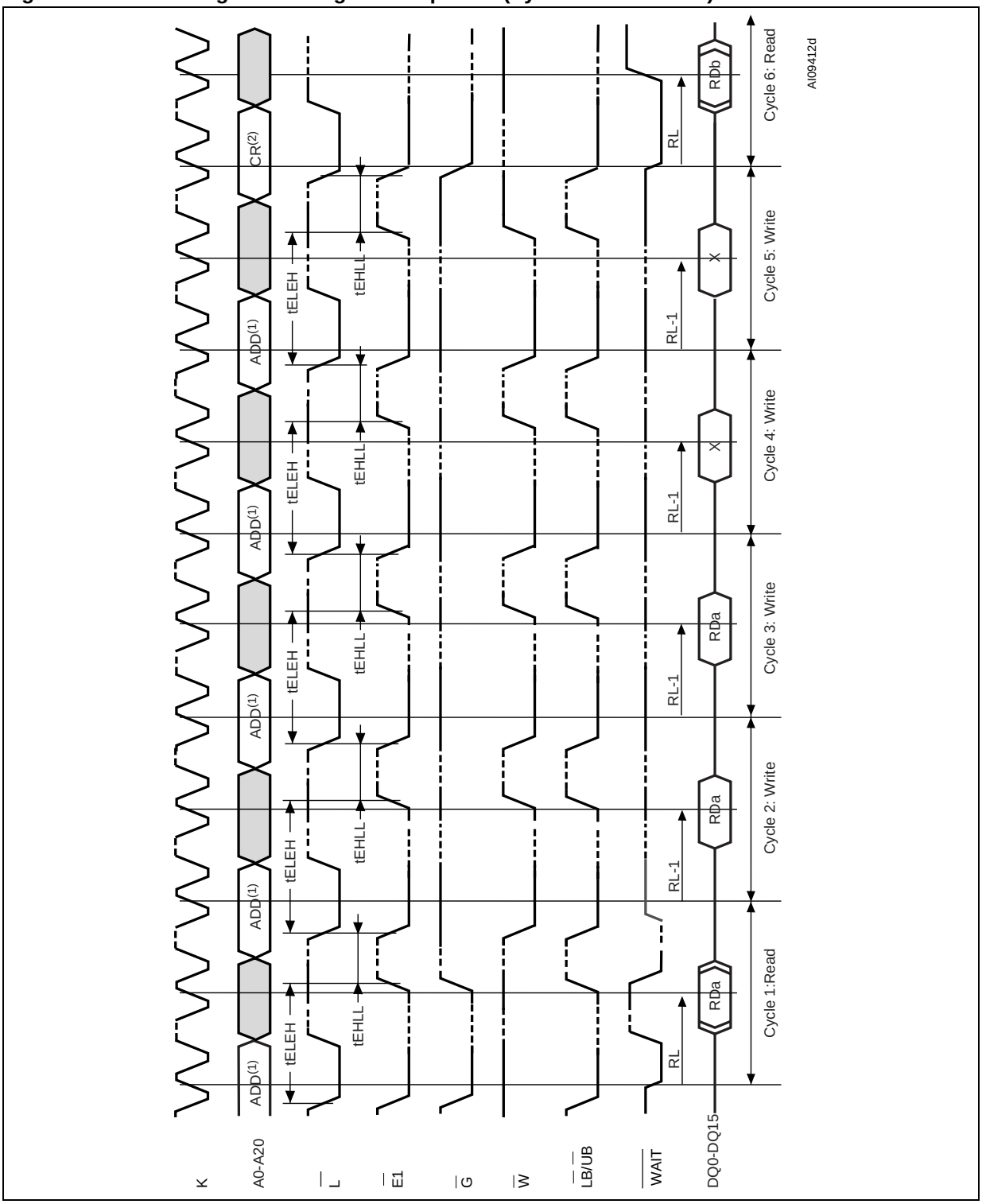
1. Both t_{EHGL} and t_{EHWL} define the minimum time to enter Standby mode.
If one of the two timing is not respected, the device enters Standby mode t_{AVAX} min after the rising edge of $\overline{E1}$.

Figure 45. Set Configuration Register Sequence (Asynchronous mode) AC Waveforms



1. All address inputs, ADD, must be High from cycle 1 to cycle 5.
2. CR is the Configuration Register settings. It must be compliant with the format specified in *Table 5: Configuration Register definition* otherwise the data programmed during the Set Configuration Register sequence may be incorrect.
3. t_{EHEL} or t_{AVAX} after the end of Cycle 6, the Set Configuration Register sequence is completed and the device returns to normal operation. t_{EHEL} and t_{AVAX} are also applicable in Synchronous mode.

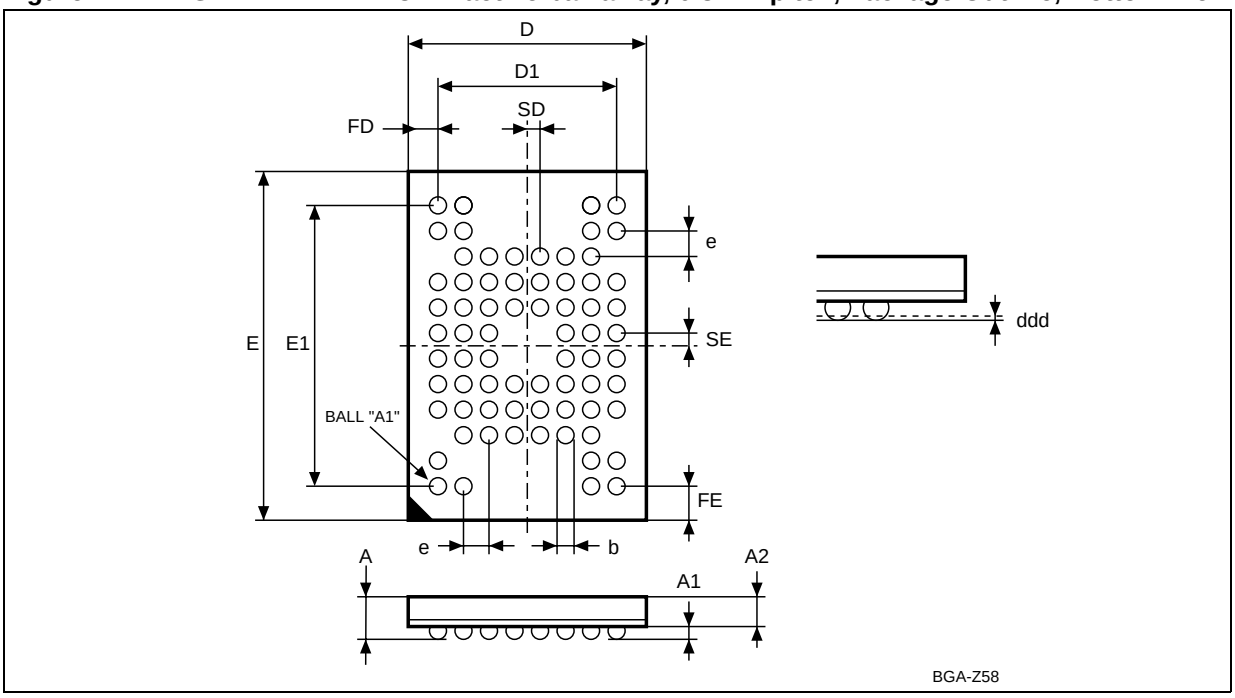
Figure 46. Set Configuration Register Sequence (Synchronous mode) AC Waveforms



1. All address inputs, ADD, must be High from cycle 1 to cycle 5.
2. CR is the Configuration Register settings. It must be compliant with the format specified in *Table 5: Configuration Register definition* otherwise the data programmed during the Set Configuration Register sequence may be incorrect.
3. t_{EHLL} after the end of Cycle 6, the Set Configuration Register sequence is completed and the device returns to normal operation.

6 Package mechanical

Figure 47. FBGA71 7x11mm - 8x12 active ball array, 0.8mm pitch, Package Outline, Bottom View



1. Drawing is not to scale.

Table 20. FBGA71 7x11mm - 8x12 active ball array, 0.8mm pitch, package mechanical data

Symbol	millimeters			inches		
	Typ	Min	Max	Typ	Min	Max
A			1.200			0.0472
A1		0.250			0.0098	
A2			0.900			0.0354
b		0.350	0.450		0.0138	0.0177
D	7.000	6.900	7.100	0.2756	0.2717	0.2795
D1	5.600			0.2205		
ddd			0.100			0.0039
E	11.000	10.900	11.100	0.4331	0.4291	0.4370
E1	8.800			0.3465		
e	0.800	–	–	0.0315	–	–
FD	0.700			0.0276		
FE	1.100			0.0433		
SD	0.400			0.0157		
SE	0.400			0.0157		

7 Part numbering

Table 21. Ordering information scheme

Example: M69AB048 B D 70 ZA 8 T

Device Type

M69 = 1T/1C Memory Cell Architecture

Mode

A = Packaged Device

K = Wafer Form

Operating Voltage

B = V_{CC} = 1.7 to 1.95V, Burst, Address/Data bus standard x16

Array Organization

048 = 32 Mbit (2M x16)

Option 1

B = 2 Chip Enable; No Write and Standby from $\overline{UB}$ and $\overline{LB}$

Option 2

D = D Die

Speed Class

70 = 70ns

Package

ZA = FBGA71, 0.8mm pitch

W = Unsawn wafer

Operative Temperature

8 = –30 to 85 °C

Shipping

T = Tape and Reel Packing

F = Lead-Free and RoHS Package, Tape and Reel Packing

The notation used for the device number is as shown in *Table 21*. For a list of available options (speed, package, etc.) or for further information on any aspect of this device, please contact your nearest STMicroelectronics Sales Office.

8 Revision history

Table 22. Document revision history

Date	Revision	Changes
15-Jan-2007	1	Initial release.

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